

A deep learning based framework for solar panel segmentation and fault classification enhanced with explainable AI

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ABSTRACT

With the increasing adoption of solar photovoltaic (PV) systems worldwide, efficient monitoring and fault detection have become essential to ensure optimal performance and longevity. Traditional PV panel inspection methods are often time-consuming, labor-intensive, and lack precision. To address these challenges, this study presents a deep learning-based framework for solar panel segmentation along with faults and abnormalities classification enhanced with Lime based Explainable AI (XAI). The proposed approach employs a hybrid segmentation model that combines the strengths of U-Net and RefineNet, leveraging DenseNet as the backbone. The segmentation module is designed to perform binary segmentation to ensure the precise localization of solar panels before the fault classification stage. For fault classification, the MobileNetV3Small-based transfer learning model has been utilized, which can classify diverse real-world solar panel fault and abnormality conditions, including physical and electrical damage, dust, bird droppings, and snow cover. Furthermore, LIME-based XAI has been incorporated to improve interpretability, providing insight into the decision-making process of the model. The segmentation model achieved an Intersection over Union (IoU) of 93.63% and 91.24% for two publicly available datasets. The classification model attained 97.53% accuracy. The proposed method outperforms existing approaches in both segmentation and classification while maintaining computational efficiency. By combining segmentation and fault classification as a sequential model, this framework enables automated solar panel monitoring with high reliability and transparency.

1. Introduction

The rapid depletion of non-renewable resources and the escalating impacts of climate change have intensified the global push toward renewable energy. As urbanization increases energy demand, the transition to cleaner sources has become vital for reducing greenhouse gas emissions and ensuring environmental sustainability. Among the available renewable energy sources, solar energy has emerged as one of the most promising alternatives due to its abundance, scalability, and ease of deployment. By 2023, global solar power capacity surpassed 1400 GW, reflecting a 35% market growth over the past decade [1,2].

Despite this rapid expansion, maintaining the performance and reliability of PV systems remained a significant challenge, particularly in large-scale and remote installations. Solar panels are prone to environmental stress due to their exposure in remote and harsh locations. Accumulated bird droppings, snow, dust, and debris can obstruct sunlight and cause localized heating, reducing efficiency and accelerating degradation. Saline air and prolonged snow exposure may corrode panel

frames and connections, while thermal cycling can induce microcracks, leading to electrical faults. Physical impacts from lightning or human activity further contribute to structural and functional damage [3].

Traditional inspection methods such as manual checks, electroluminescence testing, or thermographic imaging are time-consuming, labor-intensive, and unsuitable for continuous monitoring [4]. Although advances in computer vision and deep learning have enabled automated analysis of PV imagery, existing approaches generally address different small aspects of PV monitoring separately. Those existing deep learning based approaches either introduce solar panel segmentation and localization or fault detection separately. Moreover, most of the previous work emphasized mainly dust detection on solar panels or physical damage detection. None of the previous research considered other kinds of faults and weather related abnormalities like electrical damage, snow accumulation, and bird droppings in solar panels. Furthermore, the black-box nature of many deep learning models limits their adoption in safety-critical applications like solar panel

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monitoring, where stakeholders require transparent and interpretable predictions. There is minimal application of Explainable AI (XAI), resulting in limited interpretability and reduced trust. Also, many prior studies showed over-reliance on single datasets, limiting generalization across diverse environments.

To overcome these limitations, this study proposes an integrated deep learning framework for automated solar panel segmentation and fault/abnormality classification, enhanced with XAI. The segmentation module employs a hybrid RefineU-Net that combines U-Net and RefineNet, with DenseNet as the backbone for precise solar panel localization in aerial imagery [5]. For classification, a MobileNetV3Small-based transfer learning model detects diverse panel conditions with high accuracy and efficiency. Furthermore, LIME-based XAI is incorporated to enhance model transparency, offering interpretable insights into the classification outcomes and decision making process. The proposed framework delivers reliable, interpretable, and scalable PV system monitoring, enabling proactive maintenance strategies and supporting the sustainable expansion of solar energy infrastructure.

2. Related works

2.1. Solar panel segmentation

Early approaches of solar panel segmentation relied on traditional computer vision techniques, such as histogram thresholding, region growing, and edge detection algorithms. They showed limitations over environmental conditions like lighting, shadows, occlusion, etc [6]. Many researchers have tried to develop better segmentation models for solar panel segmentation in the past few years. Due to the growth of deep learning, Convolutional Neural Networks (CNNs) have emerged as the go-to method for segmentation tasks. CNNs have evolved significantly to meet the challenges of pixel-level classification, leading to innovations like Fully Convolutional Networks (FCNs), U-Net, DeepLab, and others.

Qingyu et al. [7] designed a framework known as ‘SolarNet+’ specifically to learn rooftop structure and orientation to figure out the potential for solar panel deployment. It had an RMSE (Root Mean Square Error) value of 19.92%. However, it has poor transferability and fails to predict properly when the roof structure changes significantly. Jonathan et al. [8] implemented FCN (Fully Convolutional Network) for semantic segmentation. It was tested on PASCAL VOC-2012 and achieved an IoU score of 62.2%. The inference time was only 175 ms. However, the ‘DeepLab’ Deep Convolutional Neural Network (DCNN) model was implemented by Liang-Chieh Chen et al. [9]. In the semantic segmentation task of the PASCAL VOC-2012 dataset, it achieved a mean Intersection over Union (mIOU) of 79.7%. But it is computationally expensive. Guosheng et al. [10] introduced a framework based on RefineNet, which managed to secure an 83.4% IoU score over the same dataset and is less computationally expensive than the previous one.

Hou et al. [11] made a comparison of DeepLabV3+, U-Net, and RefineNet over the PV dataset of PV08, PV03, and PV01. It was found that DeepLabV3+ showed better accuracy and precision in solar panel segmentation than the rest of the methods or models. However, the method showed inconsistent performance across different land types. While using DeepLabV3+, certain objects with similar spatial textures were mistakenly identified as PV modules. Additionally, the boundaries of PV modules were often either overly smoothed or exhibited a jagged, saw-toothed appearance. Rui Zhu et al. [12] attempted to solve the problem of the previous research by introducing a new model, Deep Solar PV Refiner. It combined Atrous Spatial Pyramid Pooling with a Dual-Attention Network. It also incorporates the PointRend Network. Nonetheless, these approaches encountered problems such as difficulty in detecting small and distributed PV modules, misclassification under sunlight reflection, and irregular boundaries around the detected PV areas.

Moreover, Gustavo et al. [13] developed the Mask 2 Former model for segmenting PV panels. It surpassed the performance of DeepLabV3+. It surpassed in terms of F1-score, accuracy, recall, and precision compared to prior studies. It attained an IoU score of more than 95%. Ayesha et al. [1] managed to have a good IoU score and pixel accuracy in determining various kinds of solar panel damage. They proposed a transfer learning based model where the popular VGG19 was the backbone. It is noteworthy that a class imbalance issue exists during the segmentation task since most images have background occupying the majority pixel area.

The U-Net convolutional neural network architecture excels in tasks like solar panel segmentation. Its balanced encoder–decoder framework with connecting layers allows it to capture both broad and detailed features, making it ideal for separating solar panels from intricate backgrounds. Narayana et al. [14] presented an efficient CNN named ‘MobileNet’ in the semantic segmentation of solar panels. It showed better results compared to Vanilla U-net, which is another CNN architecture primarily designed for image segmentation. Haiyang et al. [15] developed an enhanced MobileNet algorithm, which combines the strengths of both MobileNet and logistic regression methods. Arif et al. [16] proposed an architecture that combines MobileNet with U-Net to get better segmentation accuracy. However, the performance of the architecture was not verified with a diverse dataset.

Rifat et al. [17] presented a hybrid model ‘SPF-net’ with a combination of InceptionV3-Net with U-Net. It outperformed well-known CNN models, such as ResNet50, VGG16, and MobileNetV3, in predicting segmentation masks from satellite images of solar panels. Li Zhuang et al. [18] made ‘CrossNets’, which is also U-Net based. It is specifically made for small-scale residential solar panels. The paper only focused on horizontal projection areas from satellite imagery, which can cause inaccurate predictions for cases where PV modules are on slanted rooftops.

Jiafan et al. [19] proposed ‘DeepSolar’, which leverages transfer learning for training a CNN model and conducting semi-supervised segmentation. Mohamed Chahine Bouaziz et al. [4] utilized a U-Net model enhanced by image data augmentation. This technique significantly improved the model’s effectiveness by generating a larger and more diverse dataset. This enabled the model to observe images from various angles and orientations for enhanced training. It surpassed the previous work because the DeepSolar Methodology needs comparatively more computational power. It also needed a time-consuming process of labeling vast image datasets. It is required to make a supervised learning model. Identifying different types of solar panel damage is also crucial for adapting different maintenance strategies. Qi Li et al. [20] implemented Solar Diagnostics to determine all 6 levels of damage to the PV array. It achieved a Matthews Correlation Coefficient (MCC) of 0.95. The results were achieved in the pre-trained model.

2.2. Fault and abnormality classification

Solar panel classification is crucial for different types of fault detection. It can be cracks in solar panels, bird droppings, dirt and debris over solar panels, or damaged PV cells. Cracks can cause disconnections of PV cells. It can lower the power generation of the solar panels, especially diagonal and multi-directional cracks [21]. Not only that, but the presence of dust and bird droppings prevents sunlight from reaching the cells underneath [22]. Alexander et al. [23] achieved a Binary Error Rate (BER) of 7.73% in classifying defective solar panels from the other. It was done using VGG16 with some modifications. Sergiu et al. [24] implemented CNN and SVM with an accuracy of 88.42% and 82.44% in detecting solar panel faults like material defects, and degradation of cell interconnection. The latter method is multi-hardware compatible and shows constant results. One notable limitation that was revealed was the necessity of examining every solar cell manually. After comparing several pre-trained CNN models, Zeynep et al. [25] concluded that the EfficientNetB0 model had the best accuracy. It attained an accuracy

of 93.93% in detecting solar panel defects. It helped to detect cell hotspots, vegetation, soiling, etc.

A Real-time Multi-Variant Deep learning Model (RMVDM) was developed by S. Prabhakaran et al. [26]. The model can differentiate normal solar panels, panels with dust, bird droppings, broken, faded, and cracked with an accuracy of 98%. However, it requires a significant number of epochs for training, which increases computational time. Liao et al. [27] proposed a MATLAB-based approach for detecting solar panel faults from real-time images captured by an unmanned aerial vehicle (UAV). However, the study was constrained by a limited number of samples and did not account for weather-related conditions. Giovanni et al. [28] achieved an accuracy of 98% by classifying thermographic images to detect hotspots and dirt on solar panels. It was done using a CNN approach.

Poonam et al. [29] designed HyperionSolarNet, which achieved an accuracy of 96% for classification, along with an IoU score of 82% for segmenting solar panels from satellite images. However, the computational efficiency of this proposed model was not reported. P. Banda et al. [30] used GoogleNet to differentiate between two classes, defective solar panels and normal solar panels, from their electroluminescence image. It had an accuracy of 98%. However, it took 100 epochs to reach the desired result for training, and it cannot differentiate the kinds of faults present. On the other hand, Md Saif et al. [31] developed SolarNet, which gained an accuracy of 98.2% in classifying clean and dirty solar panels. It was within a training period of 30 epochs. Ayesha et al. [1] managed to detect 5 classes of solar panel damage with high classification accuracy and F1 score. Rifat et al. [17] achieved an accuracy of 94.35% while identifying different types of solar panel faults. In this research, InceptionV3-Net was combined with the U-Net architecture, which was able to differentiate faulty and dusty solar panels from clean solar panels.

2.3. Explainable AI (XAI) in solar panel fault diagnosis

Deep learning models are becoming more sophisticated. But the “black-box” nature of such models is getting much attention. People are evaluating the trustworthiness of such models. This is especially concerning in high-stakes applications such as solar energy system monitoring, predictive maintenance, and fault diagnosis. To understand the use of Explainable AI (XAI) here, the term “Explainable” refers to the ability to offer a transparent rationale for the decisions made by a model. It helps to precisely explain the underlying mechanisms or processes that produce the outputs. XAI offers insights into the predictions of deep learning (DL) models. Thus, it helps to identify the features being learned from the input to determine its classification. Thus, it increases the trust and transparency in the models [32]. XAI helps to address the issues of bias and fairness by managing the balance between bias and variance in AI and machine learning models [33].

XAI techniques are mainly of two types. One is ‘transparent’ and another is ‘post-hoc’. The second approach is more popular as it explains the model’s prediction without changing or knowing the inner workings. LIME (Local Interpretable Model-Agnostic Explanations) and SHAP (Shapley Additive Explanations) are some common examples of this approach [32]. Different works to implement explainable AI by different approaches like DeepLIFT, RISE, SHAP, etc, are quite popular [34]. SHAP is a technique for interpreting predictions made by machine learning models by assigning an importance value to each feature [35].

LIME is an approach to making a model interpretable. LIME relates between model’s prediction to the instance’s element [36]. Marco Tulio et al. [37] showed SP-LIME in their paper, where SP stands for Submodular Pick. It is a way to obtain representative and non-redundant predictions, delivering a global view of the method to observers. However, LIME has limitations. Sometimes, it can yield inconsistent results due to random perturbations and may struggle with high-dimensional

data, where local linear approximations fail to capture complex decision boundaries [38]. Purushothaman et al. [6] applied SP-LIME, a variation of LIME under XAI, to analyze the outcomes of well-known algorithms such as VGG16, DenseNet201, and others. Pavan et al. [34] used LIME explanations in the prediction analysis of the VGG16 model. However, the limitation exists in the size of the dataset used and a lack of authenticity in the labeling of the obtained data. These studies highlighted the critical role of explainable AI in improving the ability to interpret complex models. It is also essential to evaluate the trustworthiness of complex models.

2.4. Research gap and contribution

Despite significant advancements in deep learning for solar panel segmentation and fault detection, the existing research has several critical limitations.

- Most prior works focused separately on solar panel segmentation or fault classification, lacking an integrated framework that can handle both tasks in a streamlined manner. This separation resulted in inefficiencies and increased computational complexity when applied in real-world scenarios.
- Past research primarily classified faults into binary classes (clean vs. faulty) or focused on limited fault types (e.g., only dust or cracks). Real-world PV panel faults include multiple overlapping abnormalities, such as physical damage, electrical damage, dust, snow accumulation, and bird droppings, which have not been adequately addressed in a single classification model.
- Existing works showed a limited generalization across diverse environments. Models were often trained on specific single datasets, making them less adaptable to varying environments, including rooftop and utility-scale installations. There were no efficient hybrid segmentation models available to tackle these scenarios.
- None of the previous research work considered the “black box” nature of the DL model that makes it challenging to interpret why a model classifies a solar panel as faulty. These works did not incorporate Explainable AI (XAI) techniques, making their results less transparent and difficult to trust in critical decision-making scenarios.
- Existing models misclassified small-scale solar panels, including shadows, reflections, and building edges, and relied on large labeled datasets, complicating supervised learning.
- Previous research has demonstrated a dependency on big labeled datasets, hindering the deployment of supervised learning.
- Lack of study on the generalization ability of deep learning from aerial to satellite data and vice versa.

To address these research gaps, this study introduces a novel deep learning-based framework for solar panel segmentation and fault classification, enhanced with Explainable AI (XAI) for better interpretability. The primary contributions of this research include:

- This study proposes a sequential segmentation and classification module for solar panel monitoring and fault detection.
- The proposed RefineU-Net model is an improved segmentation model with a hybrid architecture. It combines the strengths of U-Net and RefineNet, utilizing DenseNet121 as a backbone to enhance feature propagation, boundary detection, and segmentation accuracy.
- Unlike resource-heavy architectures (e.g., ResNet152, EfficientNet), MobileNetV3Small is used for classification, offering a balance between low computing cost and high accuracy.
- Unlike most of the prior studies that focused only on dust or clean panels, this research introduces a multi-class classification approach. It detects six different fault and abnormality types, including physical damage, electrical damage, snow cover, dust accumulation, and bird droppings.

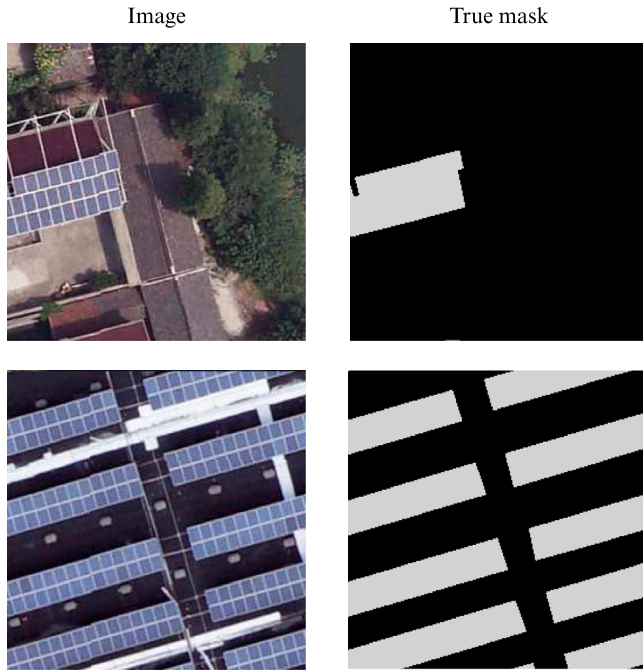


Fig. 1. Image and corresponding mask samples of PV01 dataset.

Table 1

PV01 dataset features overview.

Type	Resolution	Size (px)	No. of images
Rooftop_SteelTile	~0.1 m	256 × 256	94
Rooftop_Brick	~0.1 m	256 × 256	138
Rooftop_FlatConcrete	~0.1 m	256 × 256	413

- To enhance the interpretability of the model, Local Interpretable Model-Agnostic Explanations (LIME) has been employed. It increases transparency and trust by offering insights about how the model makes decisions. This is the first study to apply LIME to solar panel fault classification, ensuring trust and transparency in AI-based decision-making for PV monitoring.
- The approach was validated on multiple publicly available datasets, demonstrating superior Intersection over Union (IoU) for segmentation and accuracy in fault classification.
- Extensive data augmentation techniques (e.g., flipping, rotation) were incorporated to increase the robustness of the model against lighting variations, occlusions, and different PV installations.
- The proposed model was evaluated against multiple state-of-the-art models, such as U-Net, RefineNet, and DeepLabV3+ for segmentation, while EfficientNet, ResNet152, Xception, InceptionV3 for classification task.

3. Dataset and preprocessing

3.1. Segmentation dataset

The two segmentation datasets, PV01 and PV03, utilized in this research consist of satellite and aerial images collected from China's Jiangsu Province. These images cover a vast area of approximately 107,200 km² [11]. Government policies have effectively encouraged the adoption of photovoltaic systems in this area. That resulted in extensive installations in places with little land use. These installations can be found in a variety of locations, such as rooftops, saline-alkaline zones, low-density grasslands, sparse shrublands, ponds, and reservoirs.

The PV01 dataset consists of rooftop photovoltaic (PV) installations obtained through unmanned aerial vehicle (UAV) imagery with a 0.1 m

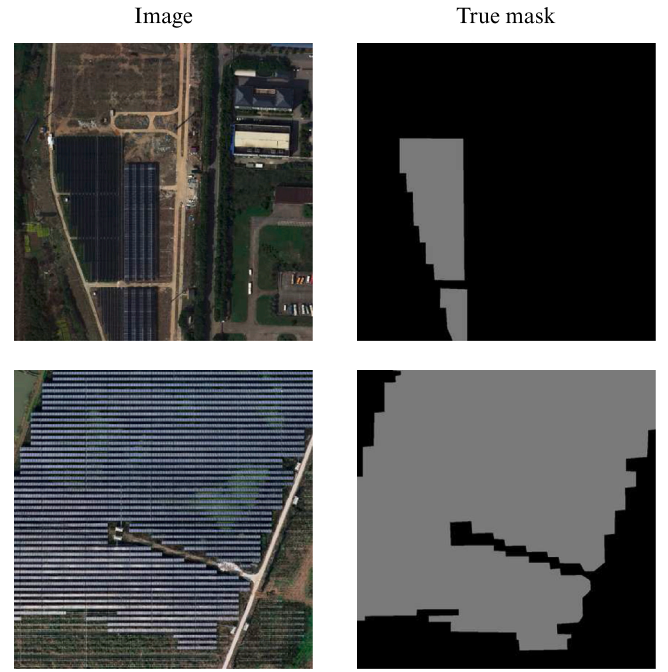


Fig. 2. Image and corresponding mask samples of PV03 dataset.

Table 2

PV03 dataset features overview.

Type	Resolution	Size (px)	No. of images
Ground_Grassland	~0.3 m	1024 × 1024	117
Ground_Shrubwood	~0.3 m	1024 × 1024	119
Rooftop	~0.3 m	1024 × 1024	236
Ground_SalineAlkali	~0.3 m	1024 × 1024	352
Ground_WaterSurface	~0.3 m	1024 × 1024	625
Ground_Cropland	~0.3 m	1024 × 1024	859

ground sampling distance (GSD). The PV03 dataset, which primarily focuses on locating distributed ground-mounted PV systems, comprises aerial photos with a GSD of 0.3 m.

In both of these datasets, each image is paired with a corresponding binary mask indicating the solar panel regions. A sample image and corresponding mask of these two datasets are represented in Figs. 1 and 2. Tables 1 and 2 provide a detailed representation of the features of these two datasets.

3.2. Classification dataset

The classification dataset contains a wide range of images representing different abnormal and fault conditions of solar panels. Dust, snow, bird droppings, and other debris build up on the surface of solar panels, reducing the efficiency of the PV modules and, hence, the amount of energy produced. Similarly, various types of physical or electrical damage can reduce the overall efficiency of the system. This dataset was utilized to examine how well the proposed model can classify the surface conditions of solar panels.

This publicly available dataset consists of 875 samples. There are six classes in this dataset: clean, dusty, snow-covered, bird-dropping, electrical damage, and physical damage. These represent both visual surface abnormalities and deeper faults affecting panel performance [39]. Fig. 3 illustrates samples of each of these classes. The dataset shows a natural imbalance across these categories. Details of this fault classification dataset are provided in Table 3.

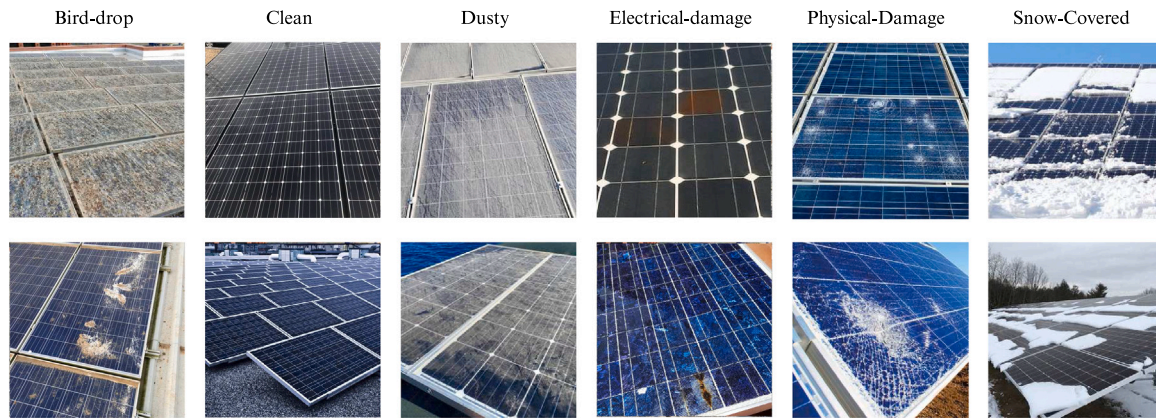


Fig. 3. Image samples of the classification dataset.

Table 3

Classification dataset overview.

Class name	No. of images
Physical-Damage	70
Electrical-Damage	104
Snow-Covered	124
Clean	194
Dusty	191
Bird-Drop	192

3.3. Data preprocessing and augmentation

Data preprocessing is a crucial step for preparing the images for practical model training. So, proper data preprocessing is necessary for both segmentation and classification datasets.

For the PV01 and PV03 datasets, all images and their corresponding mask images were resized to a fixed dimension of 256×256 pixels using bilinear interpolation. For deep learning batch processing, this consistent size is crucial. Each value was scaled down using a divisor of 255 to normalize the image pixel values to a range of 0 to 1. This scaling enhanced convergence during the training process.

The segmentation masks were transformed into a single channel, with the background's pixel values set to 0 and the solar panel areas' to 1. This was achieved using a binary thresholding technique that simplifies the mask representation.

Afterward, the images were split into training, validation, and test sets, along with the corresponding mask images, in a ratio of 80:10:10.

Different augmentation techniques were introduced to the training datasets to incorporate diversity and improve the model's robustness. Every image and its corresponding mask in the training set were arbitrarily flipped both vertically and horizontally.

The classification dataset was also split into training, validation, and test sets in an 80:10:10 ratio. To improve generalizability and prevent overfitting, the training set went under rotation and flipping-based data augmentation.

4. Proposed methodology

The proposed framework adopts a sequential modular design that integrates segmentation, classification, and explainability into an integrated solar panel monitoring system. The segmentation model localizes solar panels in aerial or satellite images, while the classification model identifies surface-level faults and abnormalities using close-range panel images. The Hybrid RefineU-Net-based segmentation model performs binary localization of solar panels (panel vs. non-panel) from aerial imagery. It generates masks that guide the classification stage by isolating panel regions and reducing irrelevant

background. The MobileNetV3small-based classification model then analyzes solar panels to detect faults (Physical and Electrical faults) and abnormalities (Dust, Bird drop, Snow cover) using image-level labels. To maintain consistency, the classifier is trained on images that mimic segmentation-guided cropping, ensuring alignment between training and deployment conditions.

Although the modules are trained independently, they operate in a sequential workflow during deployment. To enhance transparency, the classification module is complemented by a LIME-based explainable AI (XAI) component. It provided interpretable visual explanations of predictions, supporting trustworthy and actionable insights for maintenance personnel. Fig. 4 represents the overall structure of the proposed framework.

4.1. PV panel segmentation model

In this paper, a hybrid RefineU-Net model has been proposed for solar panel segmentation. This hybrid model combines DenseNet's effective feature extraction capabilities with the encoder-decoder structure of U-Net, incorporating additional refinement steps inspired by RefineNet.

The segmentation module is designed to perform binary segmentation, wherein each pixel is classified as either part of a solar panel (foreground) or a non-panel region (background). It helps to distinguish solar panel pixels from the background. Thus, this step ensures precise localization of solar panels before the fault classification stage.

4.1.1. Backbone and feature extraction

DenseNet121 serves as the backbone for the proposed segmentation model. It is popular for its efficient feature reuse and improved gradient flow. Thus, DenseNet121 facilitates better information transfer across its densely connected layers. It results in minimizing redundancy in feature learning. DenseNet121 provides robust feature extraction from the outset, utilizing the ImageNet dataset for pre-trained initialization. For this segmentation model, only the convolutional layers are utilized, with the top classification layers frozen. It allows the model to concentrate on capturing rich, deep features from the input images.

To enable multi-level feature extraction, five essential layers have been chosen from the DenseNet121 architecture. Thus, it captures both spatial and semantic details at different depths, as shown in Table 4. Each of these layers represents a distinct stage of downsampling within DenseNet, preserving both high and low-level features essential for the segmentation process. These layers' outputs act as intermediate feature maps. These are extracted through a separate downsampling model and subsequently fed into the upsampling path for reconstruction. Utilizing its pre-trained feature extraction strengths, the DenseNet121 backbone remains frozen during training, thereby reducing computational load.

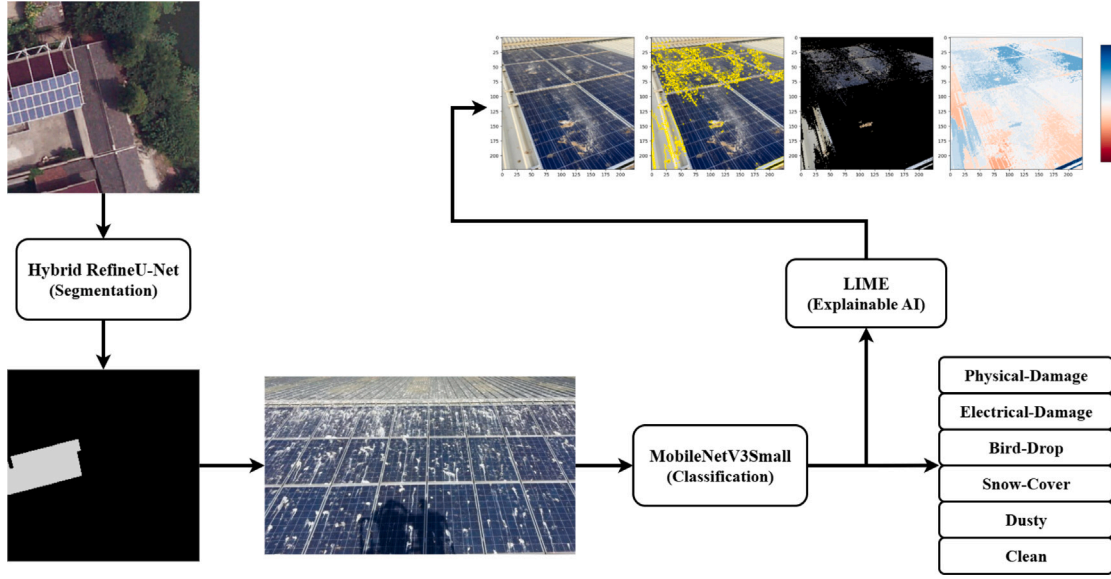


Fig. 4. Flow chart of the proposed framework.

Table 4
Feature layers of hybrid RefineU-Net model.

Layer	Resolution
conv1_relu	128×128
pool2_relu	64×64
pool3_relu	32×32
pool4_relu	16×16
relu	8×8

Table 5
Upsample layers of hybrid RefineU-Net model.

Layer	Size of input	Size of output	Filter
Upsample layer 1	8×8	16×16	256
Upsample layer 2	16×16	32×32	128
Upsample layer 3	32×32	64×64	64
Upsample layer 4	64×64	128×128	32

4.1.2. Hybrid upsampling path

The decoder part includes a progressive upsampling path designed to restore the feature maps to an input resolution of 256×256 . This upsampling structure is modeled after U-Net, with added refinement steps inspired by RefineNet.

Upsample Blocks: The upsampling path contains four blocks, each comprising a Conv2DTranspose layer with filters set to 256, 128, 64, and 32, respectively. As illustrated in Table 5, these layers gradually enhance the spatial resolution of the feature maps, transitioning from lower to higher resolutions. Each Conv2DTranspose layer employs a stride of two and a 3×3 kernel. It helps to enable spatial upsampling while maintaining coherence across feature maps. To ensure consistent weight initialization, the kernel initializer is set to a random normal distribution with a mean value of 0 and a standard deviation of 0.02.

Normalization and Dropout: After every transposed convolution layer, batch normalization has been used to improve generalization and encourage stable training. A dropout layer with a rate of 0.2 has been added during training to reduce overfitting. This process randomly deactivates neurons as a form of regularization.

Skip Connections: Following a U-Net-like architecture, skip connections link corresponding encoder layers as well as decoder paths, which helped to recover spatial features lost during downsampling. The encoder's (DenseNet121) feature maps are concatenated with the

decoder's upsampled maps. This enables the model to utilize both high-level contextual information and detailed spatial features, enhancing segmentation performance.

4.1.3. Feature refinement and multi-scale fusion

Although the overall model's structure resembles U-Net, it includes refinement steps influenced by RefineNet. Following each feature concatenation between the encoder and decoder, a 3×3 conv2D layer is used to enhance the combined feature maps. This step adds non-linearity and improves the integration of features across various resolution levels, aligning with RefineNet's focus on multi-scale fusion. ReLU activations have been used in the refinement process to add non-linearity. It enhances learning capacity and aids in recovering fine spatial details from the feature maps. This hybrid design thus combines U-Net's effective upsampling capabilities with RefineNet-inspired refinement across multiple architectural levels.

4.1.4. Final output layer

In the hybrid RefineU-Net model, a Conv2DTranspose layer has been used as the final layer. This layer converts the last upsampled feature map of size 128×128 to the target output size of 256×256 . This layer uses a 3×3 kernel with a stride of 2. As it is a binary segmentation task, a sigmoid activation function is implemented. As a result, the output is a single-channel image where each pixel reflects the possibility that it belongs to the solar panel.

4.1.5. Training strategy

The proposed model utilizes the Adam optimizer for its ability to adapt learning rates and efficiently minimize loss. This optimizer is particularly suitable for sophisticated models and large datasets. A Model Checkpoint mechanism has been implemented to ensure optimal performance by saving the best-performing model. Additionally, an Early Stopping callback has been employed to track validation loss and halt training if there is no improvement for five epochs. Thus, it prevents the risk of overfitting.

To address class imbalance in this segmentation task, a combined loss function has been implemented. Dice Loss and Binary Cross-Entropy (BCE) are integrated into this combined loss function. Combined loss prioritizes the overlap between the ground truth and predicted masks. Thus, it improves segmentation accuracy. The Combo Loss function has been represented in Eq. (1).

$$\text{Combo Loss} = \eta \cdot \text{BCE} + (1 - \eta) \cdot \text{Dice Loss} \quad (1)$$

where the weighting factor η balances the contributions of Dice Loss and BCE. The mathematical definition of BCE is presented in Eq. (2). The ground truth is represented by t_i , the number of pixels is N , and every pixel has a predicted probability denoted as p_i .

$$\text{BCE} = -\frac{1}{N} \sum_{i=1}^N [t_i \log(p_i) + (1 - t_i) \log(1 - p_i)] \quad (2)$$

Dice Loss is formulated in Eq. (3), where ϵ represents a negligible value added to avoid division by zero.

$$\text{Dice Loss} = 1 - \frac{2 \sum_{i=1}^N t_i p_i + \epsilon}{\sum_{i=1}^N t_i + \sum_{i=1}^N p_i + \epsilon} \quad (3)$$

4.2. Fault and abnormality classification model

Different transfer learning models, such as DenseNet, ResNet, VGG, and InceptionV3, can be employed for detecting faults and abnormalities in solar panels. In this paper, a MobileNetV3Small-based transfer learning model has been implemented for the classification task.

4.2.1. Model architecture and transfer learning details

The lightweight nature and high computational efficiency of MobileNetV3Small make it particularly suitable as the backbone of the fault and abnormality classification model. The architecture MobileNetV3Small includes several key components, such as:

- **Depthwise Separable Convolutions:** By carrying out depthwise and spatial convolution independently, these layers minimize computation [40].
- **Inverted Residual Blocks:** An initial expansion, depthwise separable convolution, and a projection layer are features of these inverted residual blocks. It maintains necessary features as well as efficiency. The inverted residual block can be expressed as Eq. (4).

$$u = r + F(r) \quad (4)$$

Here, the input is r , $F(r)$ represents the series of transformations (e.g., activation and convolution), and u is the output. To enable effective gradient flow through the network, $F(r)$ is added to the original input if a residual link is present.

- **Hard-Swish Activation Function:** This function offers better non-linearity at a lower computing cost when compared to standard activation functions. It improves efficiency and performance by combining a linear and a non-linear function. MobileNetV3Small's hard-swish activation can be expressed as Eq. (5).

$$h_{\text{swish}}(r) = r \cdot \frac{\text{ReLU6}(r + 3)}{6} \quad (5)$$

MobileNetV3Small is pre-trained on the ImageNet dataset. The model can initialize with pre-trained weights that allow it to grasp the features learned from a large-scale dataset. During training, the convolutional layers remain frozen, preserving essential low- and mid-level feature extraction capabilities, such as edge and texture detection. This approach focuses training on the classification layers specific to the solar panel dataset. Thus, it reduces the risk of overfitting to this relatively small dataset. By freezing the convolutional layers, the model focuses on learning relevant features rather than fitting to noise in the dataset.

4.2.2. Transfer learning adaptation

To adapt the pre-trained MobileNetV3Small for solar panel fault and abnormality detection, additional layers have been added:

- **Flatten Layer:** The outcome of the convolutional base is flattened and turned into a single vector, converting the multi-dimensional output into a form suitable for fully connected layers.

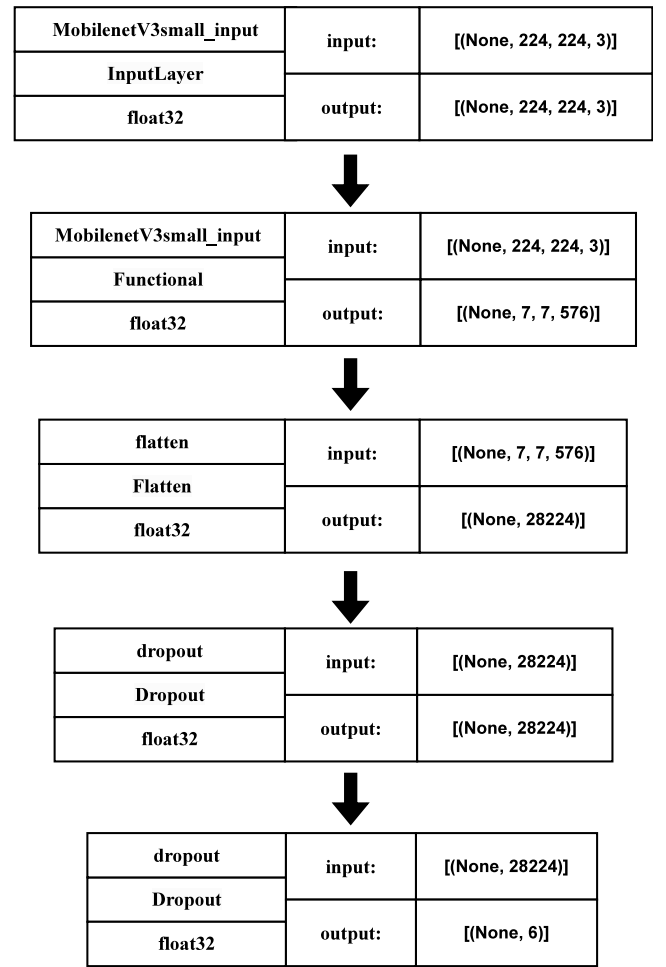


Fig. 5. Complete architecture of the classification model.

- **Dropout Layer:** A Dropout layer having a rate of 0.2 has been introduced to reduce overfitting. This layer mitigates overfitting by deactivating 20% of the neurons at random during training, encouraging the model to generalize better.

$$b_i = \begin{cases} 0, & \text{with probability } p = 0.2 \\ a_i, & \text{with probability } 1 - p \end{cases} \quad (6)$$

- **Dense Layer:** A Dense layer with six output units and a softmax activation function has been included for classification. The six units correspond to the six possible solar panel conditions that can be represented as Eq. (7).

$$\hat{q}_n = \frac{\exp(j_n)}{\sum_{i=1}^6 \exp(j_i)}, \quad n = 1, \dots, 6 \quad (7)$$

where j_n represents the logit score for class n . The softmax function produces a probability distribution across all possible classes, enabling multi-class classification.

The complete architecture of the classification model is illustrated in Fig. 5.

4.2.3. Training configuration and hyperparameters

In this classification model, the Adam optimizer has been implemented for its ability to handle sparse gradients effectively. Its adaptive learning rates contribute to faster and more stable convergence during training.

$$m_t = \gamma_1 m_{t-1} + (1 - \gamma_1) p_t \quad (8)$$

$$v_t = \gamma_2 v_{t-1} + (1 - \gamma_2) p_t^2 \quad (9)$$

$$\beta_t = \beta_{t-1} - \alpha \frac{\hat{r}_t}{\sqrt{\hat{u}_t} + \epsilon} \quad (10)$$

where p_t represents the gradient, r_t and u_t are first and second-moment estimates, α is the learning rate, and ϵ is a small value added to avoid division by zero. As a loss function, Sparse categorical cross-entropy has been used. Since the classes are represented as integer labels rather than one-hot encoded vectors, Sparse categorical cross-entropy was well suited for this classification.

$$L(o, \hat{o}) = - \sum_{j=1}^N o_j \log(\hat{o}_j) \quad (11)$$

where o_j indicates the true label and $\hat{o}_j$ represents the predicted probability for class j .

To improve the model training process, ModelCheckpoint and EarlyStopping callbacks have been implemented. The configuration of ModelCheckpoint is such that it saves model weights whenever a decrease in validation loss is detected. An early stopping mechanism has been established with a patience threshold of five epochs. Training ceases if the validation loss fails to improve during five consecutive epochs. This mitigates overfitting and optimizes the utilization of training resources.

5. Explainable AI (XAI)

5.1. Need for interpretability

The integration of Artificial Intelligence (AI) models in solar panel fault detection underscores a critical need for interoperability.

Stakeholders, including engineers and maintenance staff, frequently encounter inquiries about the reliability of predictions. This is especially critical in high-stakes scenarios when erroneous forecasts may lead to financial detriment or safety hazards. Despite the impressive accuracy of sophisticated machine learning models, especially deep learning architectures, their inherent “black-box” nature makes it difficult to grasp their reasoning, which may prevent timely interventions [33].

Recent advancements in Explainable AI (XAI) provide frameworks for interpreting complex models to address these challenges. XAI improves transparency and fosters trust by addressing bias and fairness in AI and ML algorithms.

5.2. Local Interpretable Model-Agnostic Explanations (LIME)

LIME is an innovative technique designed to interpret the predictions generated by any classifier. Lime explains any classifier's prediction by a locally interpretable model constructed around the prediction [37]. Essentially, LIME makes small perturbations to input data locally to observe how predictions are affected. Local perturbations focus on explaining a single prediction for a specific instance, while global perturbations examine the model's behavior across the entire dataset. Fig. 6 represents the overall working structure of the LIME algorithm.

In the mathematical formulation of the LIME explainer, $f : \mathbb{R}^d \rightarrow \mathbb{R}$ represents the model that needs to be explained. The main representation of the instance under explanation is $x \in \mathbb{R}^d$. The possibility (or binary indicator) of an instance x belonging to a particular class is represented by $f(x)$ in a classification context.

Furthermore, $\pi_x(y)$ defines the locality surrounding x . It is a proximity estimation between an instance x and y . The accuracy with which the explanation function g approximates f inside the locality denoted by π_x is represented by the loss function $L(f, k, \pi_x)$. For locally weighted squared loss, the loss function can be represented as Eq. (12).

$$L(f, k, \pi_x) = \sum_{y, y_0 \in Y} \pi_x(y) \cdot (f(y) - k(y_0))^2 \quad (12)$$

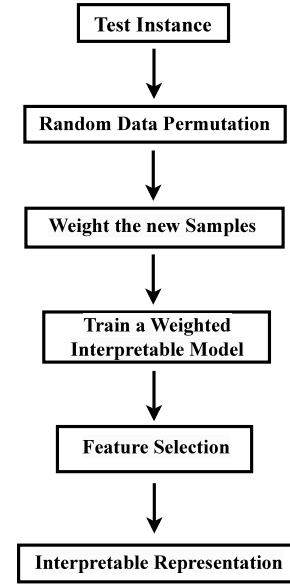


Fig. 6. Flowchart of LIME.

where $f(y)$ denotes the labels derived from the original model and Y represented a dataset with sampled instances y . In order to generate the initial instance representation, the term y_0 functions as a binary indicator for perturbed samples. The function $k(y_0)$ is typically chosen from a class of linear models, specifically $k(z_0) = w_k \cdot z_0$. By highlighting the importance of each sampled case based on its proximity to the original occurrence x in the feature space, this formulation encapsulates the core of the locally weighted squared loss. The term π_x acts as a locality measure and can be formulated as Eq. (13).

$$\pi_x(y) = \exp \left(- \frac{Q(x, y)^2}{\delta^2} \right) \quad (13)$$

Here, the distance metric (such as the cosine or Euclidean distance) between instances x and y is denoted by Q . One parameter that needs to be carefully chosen is the kernel bandwidth δ . LIME determines the explanation by solving the optimization problem mentioned in Eq. (14).

$$\xi(x) = \arg \min_{k \in K} L(f, k, \pi_x) + \Omega(k) \quad (14)$$

Here, $\xi(x)$ denotes the complex model f 's local behavior around instance x . In this equation, $L(f, k, \pi_x)$ assesses how closely the interpretable model k approximates the black-box model f for the perturbed sample π_x , while $\Omega(k)$ measures the complexity of the interpretable model.

The objective is to balance interpretability and reliability by reducing the loss function $L(f, k, \pi_x)$ while keeping the explanation complexity $\Omega(k)$ at a level understandable to humans. To achieve this, the explanation function k is penalized when it becomes excessively complex, keeping the surrogate model within reasonable complexity bounds. Thus, LIME helps to improve the comprehension of its decision-making process.

LIME is applied as a post hoc interpretability method after the model has been trained. It helps to analyze individual predictions made by the classifier. This allows users to inspect how local regions of input images influence specific predictions, without affecting the model's underlying structure or training dynamics.

6. Evaluation metrics

While evaluating the performance of the proposed model, selecting appropriate metrics is crucial. For the segmentation model, IoU was used as the primary assessment metric, along with the F1 score, accuracy, precision, recall, and the Dice coefficient. For evaluating the fault and abnormalities classification algorithm, the F1 score, MCC score, accuracy, precision, and recall were assessed.

The Intersection over Union (IoU): IoU is a critical metric for assessing the performance of semantic segmentation models, particularly because it addresses class imbalance. It calculates the overlap between the predicted segmentation of the solar panel area and the corresponding ground-truth region. The IoU score can be mathematically formulated as Eq. (15) where TP and TN denote true positives and true negatives. Whereas FP and FN represent false positives and false negatives.

$$\text{IoU} = \frac{\text{TP}}{\text{TP} + \text{FN} + \text{FP}} \quad (15)$$

Dice coefficient: This metric assesses the segmentation quality by comparing the similarity between the predicted output and the actual ground truth. The dice coefficient ranges from 0 to 1, where 1 indicates perfect overlap, and 0 indicates no overlap. Eq. (16), represents the mathematical formulation of the Dice coefficient.

$$D = \frac{2|X \cap Y|}{|X| + |Y|} \quad (16)$$

where $|X \cap Y|$ represents the common pixels between prediction and ground truth. $|X|$ and $|Y|$ denote the number of pixels in each respective segmentation.

Accuracy: Accuracy is a widely used evaluation metric that calculates the ratio of correctly classified instances. It indicates the overall effectiveness of the model in both segmentation and fault classification. In solar panel segmentation, it reflects how well the predicted segments match the actual panel areas. In fault and abnormality detection, accuracy shows the extent to which the model correctly classifies different faults and abnormalities as represented in Eq. (17).

$$\text{Accuracy} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}} \quad (17)$$

Precision: Precision measures the proportion of correctly predicted positive instances out of all positive predictions as represented in Eq. (18). Precision in solar panel segmentation refers to the degree to which the segmented areas match the actual solar panel areas, thereby reducing the inclusion of non-solar panel areas. In fault detection, high precision implies that most of the identified faults and abnormalities are indeed correct.

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \quad (18)$$

Recall (Sensitivity): As formulated in Eq. (19), the percentage of accurately predicted positive instances among all actual positive instances is known as recall. In segmentation, it assesses how well the model can capture every relevant solar panel part in the image. In fault detection, it assesses how well the model predicts every instance of faults and abnormalities in solar panels.

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}} \quad (19)$$

F1 Score: The F1 Score is the harmonic mean of precision and recall as expressed in Eq. (20). This score signifies the model's capability to predict positive instances without false positives and negatives. In solar panel segmentation, the F1 Score captures the capacity of a model to segment panels without under-segmentation or over-segmentation. In fault detection, there is a trade-off between accurately detecting faults (recall) and ensuring that the detected faults are accurate (precision).

$$F1 = \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \quad (20)$$

Table 6

Experimental setup.

Component	Details
Operating system	Ubuntu 20.04 (Kaggle Environment)
Programming language	Python 3.10
Hardware CPU	Intel Xeon 2 vCPUs @ 2.20 GHz
RAM	13 GB
GPU	NVIDIA Tesla T4 (16 GB)
Software framework	TensorFlow v2.15.0, Keras
IDE/Platform	Kaggle Jupyter Notebook

Matthews Correlation Coefficient (MCC): As formulated in Eq. (21), MCC is a crucial metric for evaluating fault detection models. It offers a holistic evaluation of model performance by accounting for true positives, true negatives, false positives, and false negatives. MCC score ranges from -1 to $+1$. Here, -1 signifies total inverse predictions, $+1$ indicates perfect predictions, and 0 represents random performance.

$$\text{MCC} = \frac{\text{TP} \times \text{TN} - \text{FP} \times \text{FN}}{\sqrt{(\text{TP} + \text{FP})(\text{TP} + \text{FN})(\text{TN} + \text{FP})(\text{TN} + \text{FN})}} \quad (21)$$

When the model's MCC score is high, it indicates that it is performing well across all classes. Thus, it is not dominated by class imbalances, which is important for fault classification.

7. Result and analysis

This experiment was conducted using Kaggle's cloud-based computational environment. A detailed overview of the experimental environment is presented in Table 6, which clearly outlines the hardware and software configurations used in this study.

7.1. Evaluation of segmentation model

The hybrid RefieU-Net model's segmentation performance and result analysis for the PV01 and PV03 datasets are presented in this section.

7.1.1. Training analysis

Figs. 7 and 8 present the training and validation loss curves, along with the IoU curve for the hybrid RefineU-Net model applied to the PV01 and PV03 datasets, respectively. During the training process, the model exhibits efficient convergence on both datasets. For the PV01 dataset, the loss decreases steadily with minimal fluctuations, indicating a stable learning process that minimizes overfitting. Over 24 epochs, the validation IoU increases from 38.30% to 93.41%, while the validation loss progressively decreases from 0.788 to 0.023.

For the PV03 dataset, the validation curves show some fluctuations in the early epochs. However, they become more consistent as training progresses. Within 19 epochs, the loss drops from 0.272 to 0.057, while the validation IoU rises from 61.75% to 91.43% for this dataset.

Overall, the model converges in a reasonable number of epochs for both datasets.

7.1.2. Performance of segmentation model using PV01 dataset

Table 7 represents the performances of the proposed hybrid RefineU-Net model compared to other state-of-the-art algorithms for PV01.

These results correspond to a binary segmentation task where each pixel is classified as either solar panel (foreground) or background. Thus, except for the macro and weighted F1 score, the reported IoU and other performance metrics are computed with respect to the foreground class (solar panel), which is the primary target of interest.

The hybrid RefineU-Net achieved an IoU of 93.63% with an Accuracy of 98.3%. Whereas other segmentation models like U-Net, RefineNet, and DeeplabV3+ provided an IoU score of 88.77%, 89.65%,

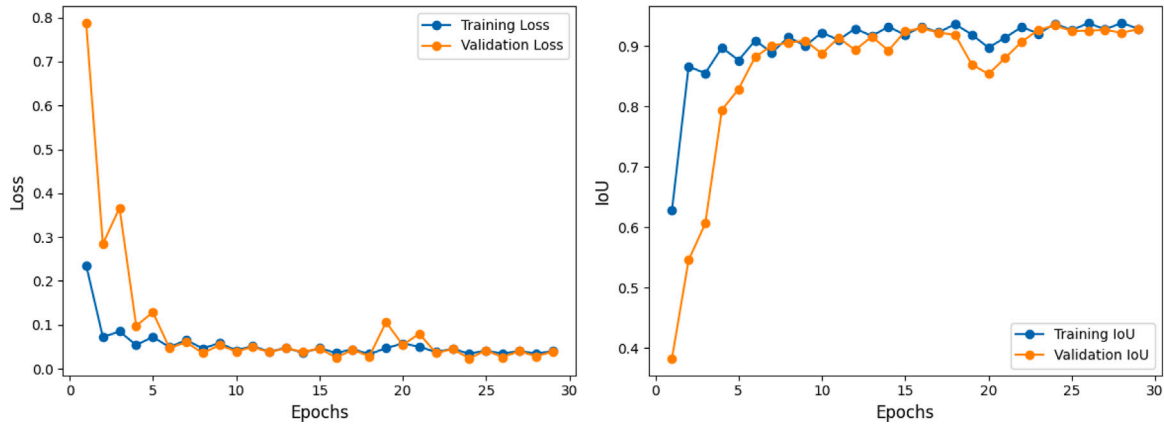


Fig. 7. Loss and IoU curve for PV01 dataset.

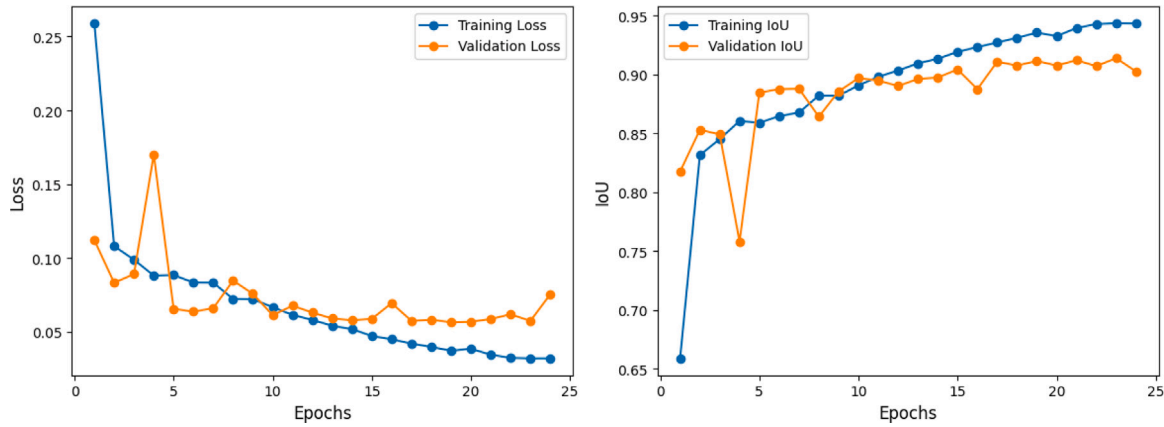


Fig. 8. Loss and IoU curve for PV03 dataset.

Table 7

Performance analysis for PV01 dataset.

Model	IoU	Accuracy	Dice coefficient	Recall	Precision	F1 score	F1 (macro)	F1 (weighted)
U-Net	88.77	96.67	87.94	95.69	93.87	94.76	96.47	97.21
RefineNet	89.65	96.75	91.56	96.22	94.1	95.14	96.56	97.16
DeepLabV3+	92.2	97.88	95.94	96.03	96.57	96.29	97.55	98.07
Hybrid RefineU-Net	93.63	98.31	96.7	97.45	97.04	97.24	97.97	98.25

Table 8

Performance analysis for PV03 dataset.

Model	IoU	Accuracy	Dice coefficient	Recall	Precision	F1 score	F1 (macro)	F1 (weighted)
U-Net	87.16	96.71	85.62	93.25	94.31	93.64	95.74	96.64
RefineNet	87.36	96.93	93.17	92.34	96.06	94.08	95.93	96.75
DeepLabV3+	88.38	97.57	93.97	93.67	96.28	94.88	96.39	97.10
Hybrid RefineU-Net	91.24	98.06	95.37	95.02	96.95	95.93	97.08	97.67

and 92.2%, respectively. Thus, the proposed model significantly outperforms other models with an IoU improvement of 2%–5%. This outperforming performance was observed across all other metrics. The value of the Dice Coefficient obtained is 96.48% for RefineU-Net, demonstrating more precise boundary detection compared to DeepLabV3+ (95.94%), RefineNet (91.56%) and U-Net (87.94%).

The proposed model's superior Recall value of 97.45% demonstrates its efficacy in accurately identifying panel regions. It also minimizes the likelihood of false negatives. This is particularly crucial in situations where the absence of a covered space for solar panels could result in inadequate maintenance of the panels. Furthermore, the model's F1 Score of 97.24% highlights its robustness and appropriate precision–recall balance.

In addition, the inclusion of F1 (macro) and F1 (weighted) provides a more comprehensive evaluation by addressing class imbalance in segmentation tasks. The hybrid RefineU-Net achieved the highest F1 (macro) of 97.97% and F1 (weighted) of 98.25%, outperforming DeepLabV3+, RefineNet, and U-Net. These results demonstrate that the model is not biased toward dominant classes (such as background pixels). It provides consistent performance across all pixel categories, confirming its robustness in real-world deployment scenarios.

7.1.3. Performance of segmentation model using PV03 dataset

Table 8 outlines the performance results for the PV03 dataset. The hybrid RefineU-Net outperformed all other segmentation models on every evaluation metric, similar to the PV01 dataset. It outperformed

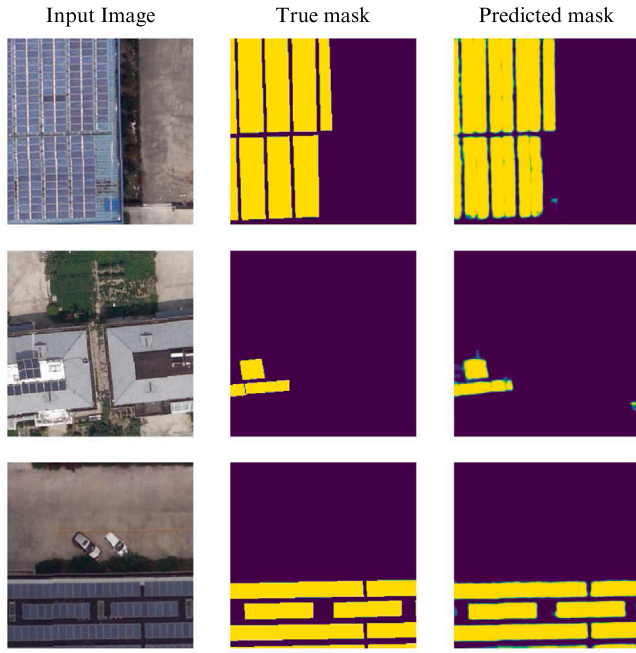


Fig. 9. Solar panel image with corresponding true mask and predicted mask from PV01 dataset.

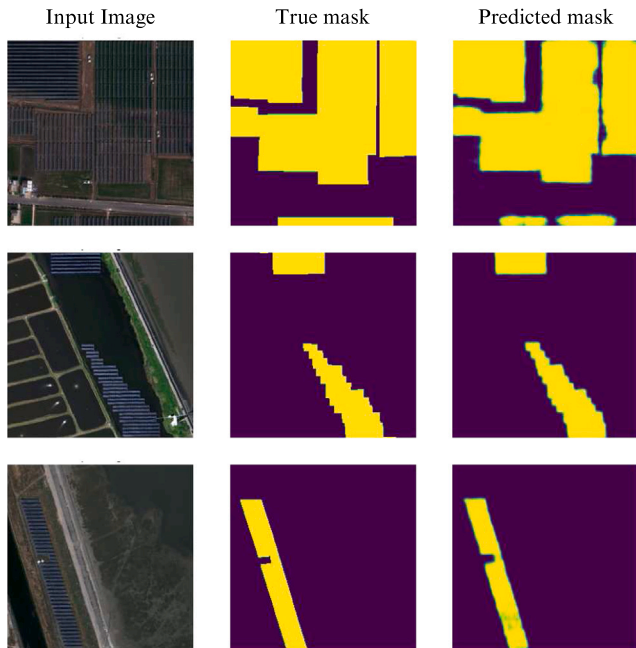


Fig. 10. Solar panel image with corresponding true mask and predicted mask from PV03 dataset.

U-Net (87.16%), RefineNet (87.36%), and DeepLabV3+ (88.38%) with an IoU score of 91.24%. This 3%–4% increase in IoU over the baseline models demonstrates how well the model generalizes to challenging scenarios with complex panel patterns and erratic lighting.

The hybrid RefineU-Net model demonstrated its efficacy in pixel-wise classification with an outstanding accuracy of 98.06%. The model achieved a recall rate of 95.63% and a precision of 95.17%, effectively managing false negatives and reducing false positives. The model's robust segmentation performance is underscored by its F1 score of 95.33%, reflecting an equitable balance between recall and precision.

Table 9

Performance analysis on PV01 for different backbones.

Backbone	IoU	Accuracy	Dice	Recall	Precision	F1
MobileNetV3	57.42	88.88	72.73	81.67	76.74	78.84
ResNet50	64.57	90.04	78.36	83.25	84.02	83.51
InceptionV3	73.71	92.93	84.79	90.63	87.93	89.23
VGG16	90.65	97.40	94.98	94.62	96.81	95.60
Xception	92.71	97.93	96.19	96.74	96.80	96.77
DenseNet121	93.63	98.31	96.7	97.45	97.04	97.24

Table 10

Performance analysis on PV03 for different backbones.

Backbone	IoU	Accuracy	Dice	Recall	Precision	F1
MobileNetV3	69.79	91.77	81.87	86.53	86.59	86.06
ResNet50	70.46	92.54	82.24	82.36	90.87	85.91
InceptionV3	84.13	96.37	91.32	93.95	93.16	93.51
VGG16	87.57	97.11	93.06	95.14	93.87	94.27
Xception	88.93	97.56	93.97	95.42	94.69	94.90
DenseNet121	91.24	98.06	95.37	95.02	96.95	95.93

Table 11

Performance analysis on PV01 for different loss functions.

Loss function	IoU	Accuracy	Dice	Recall	Precision	F1
Class weighted BCE	93.21	97.99	96.47	96.89	96.41	96.64
Pixel weighted BCE	92.94	97.88	96.32	97.11	95.75	96.41
Focal loss	92.77	98	96.24	96.17	97.23	96.69
Combo loss	93.63	98.31	96.7	97.45	97.04	97.24

Table 12

Performance analysis on PV03 for different loss functions.

Loss function	IoU	Accuracy	Dice	Recall	Precision	F1
Class weighted BCE	87.95	97.16	93.51	97.26	92.70	94.86
Pixel weighted BCE	83.84	96.2	91.07	98.35	88.89	93.28
Focal loss	69.65	96.91	81.95	91.99	97	94.38
Combo loss	91.24	98.06	95.37	95.02	96.95	95.93

Similarly, for this dataset, the proposed approach provided better macro and weighted F1 scores with 97.08% and 97.67%, respectively. It ensures that the model can deliver stable performance even if there is a class imbalance.

7.1.4. Performance visualization

Figs. 9 and 10 illustrate how accurately hybrid RefineU-Net can segment the boundaries of solar panels for the PV01 and PV03 datasets, respectively. The predicted masks closely resembled the ground truth in most areas. This demonstrates the model's ability to capture complex geometry accurately, even in the presence of occlusions from objects or shadows.

7.1.5. Ablation study

Backbone comparison. Tables 9 and 10 represent the performance comparison of the RefineU-Net model for different encoder backbones. Across both PV01 and PV03 datasets, the proposed DenseNet121 backbone consistently outperformed all other backbones. It achieved the highest IoU 93.63% on PV01, 91.24% on PV03 datasets. The dense connectivity pattern of DenseNet121 with feature reuse and richer gradient flow ensures this superior performance. It ensures improved delineation of panel boundaries and small-scale features as the most effective backbone for the proposed framework.

Impact of different loss functions. Tables 11 and 12 summarize the segmentation performance of the proposed RefineU-Net model on the PV01 and PV03 datasets using various loss functions. For both of the datasets, the Combo Loss, which integrates Dice and Binary Cross-Entropy components, consistently surpassed the other loss functions across all key metrics. On PV01, Combo Loss achieved an IoU of

Table 13

Performance Comparison with existing work for PV01 dataset.

Model	IoU	Accuracy	Dice	Recall	Precision	F1
Jiang et al. [11]	86.80	98.30	–	89.40	92.80	91.10
Zhu et al. [12]	87.60	95.84	–	95.84	91.06	93.39
This paper	93.63	98.31	96.7	97.45	97.04	97.24

Table 14

Performance Comparison with existing work for PV03 dataset.

Model	IoU	Accuracy	Dice	Recall	Precision	F1
Rudro et al. [17]	76.31	95.63	83.41	–	–	–
Jiang et al. [11]	90.80	98.30	–	93.10	95.90	94.50
This paper	91.24	98.06	95.37	95.02	96.95	95.93

Table 15

Performance analysis for validation set.

Model	Accuracy	Precision	Recall	F1	MCC
NASNetMobile	55.21	62.88	50.63	51.92	45.53
InceptionV3	62.50	67.20	62.17	63.58	56.59
DenseNet	70.31	73.56	70.78	71.59	65.76
Xception	74.48	76.03	74.19	74.53	69.66
RegNetY320	83.33	87.83	80.84	82.78	81.31
VGG19	80.21	83.63	80.99	81.81	78.07
EfficientNetV2B0	88.02	91.05	87.78	89.17	86.83
ResNet152	93.75	94.18	95.21	94.39	92.42
ConvnextXlarge	88.28	89.98	89.78	89.75	85.73
EfficientNetV2B3	93.75	95.78	92.26	93.24	92.24
MobileNetV3Small	97.92	97.89	98.72	98.24	97.44

93.63%. It reflects improved overlap between predicted masks and ground truth compared to Class Weighted BCE (93.21%) and Focal Loss (92.77%). The Dice coefficient and F1-score improvements further demonstrate enhanced boundary delineation and class balance handling. Similarly, on PV03, Combo Loss outperforms other loss functions with a greater margin. The combo loss provided an IoU improvement of 3% to 20% over other loss functions. This superior performance can be attributed to the Combo Loss's ability to simultaneously optimize pixel-wise accuracy and region-based similarity.

7.1.6. Comparison with existing works

Table 13 presents the performance of the hybrid RefineU-Net model for the PV01 dataset compared to earlier studies. Jiang et al. [11] presented a segmentation technique based on DeepLabV3+, which achieved an F1 score of 91.10% with an IoU score of 86.80%. Zhu et al. [12] proposed an approach employing U-Net, which achieved an IoU of 87.60%. On the contrary, the hybrid RefineU-Net model presented in this paper outperformed these earlier techniques, demonstrating a 6%–7% improvement with an IoU score of 93.63%.

Similarly, the proposed method demonstrated a more balanced performance for the PV03 dataset, compared to previous research works, as detailed in Table 14. The same method was used for both the PV01 and PV03 datasets by Jiang et al. [11], who obtained an IoU of 90.80% for PV03. On the other hand, Rudro et al. [17] presented a different method using U-Net with MobileNet backbone, but it underperformed, obtaining a Dice Coefficient of 83.41% and an IoU of 76.31%. In comparison, the proposed model showed an improved IoU score of 91.24%. Additionally, it demonstrated balanced performance in terms of F1 score, recall, and precision.

Overall, the proposed approach demonstrated strong segmentation performances across various datasets.

7.2. Evaluation of classification model

This section presents a comprehensive evaluation of the proposed MobileNetV3small-based classification model.

Table 16

Performance analysis for test set.

Model	Accuracy	Precision	Recall	F1	MCC
NASNetMobile	53.09	58.12	47.20	47.74	41.21
InceptionV3	68.52	76.67	70.95	72.16	66.74
DenseNet	69.14	71.16	69.41	69.77	63.67
Xception	69.14	72.66	68.47	69.40	63.77
RegNetY320	79.63	86.28	78.61	80.99	77.56
VGG19	82.10	84.67	83.16	83.69	80.10
EfficientNetV2B0	87.65	91.13	87.42	88.60	86.47
ResNet152	91.36	91.05	89.66	90.25	89.02
ConvnextXlarge	89.95	90.75	89.70	89.97	87.73
EfficientNetV2B3	92.59	94.12	92.34	93.33	90.28
MobileNetV3Small	97.53	97.74	97.74	97.74	96.93

7.2.1. Training process

Fig. 11 illustrates the training and validation loss and accuracy curves for the proposed model. During the initial epochs, a significant improvement in the validation loss and accuracy can be observed. In the first four epochs, the validation loss decreased to 0.59 from 1.87, and the accuracy increased from 70.21% to 92.71%. This indicates effective learning in the early stages. This improvement of validation loss and accuracy continued, and the best result was observed in 14 epochs with a validation accuracy of 97.92% and a validation loss of 0.15. The training continued for a few more epochs until the early stopping callback halted the process to prevent overfitting.

7.2.2. Performance on validation set

Table 15 provides a comprehensive comparison of the proposed model with other classification models on the validation set. The validation results highlight that MobileNetV3Small consistently outperformed other deep learning architectures. While models such as EfficientNetV2B3 and ResNet152 exhibited competitive accuracy, they required significantly higher computational resources, making them less practical for real-time deployment. In contrast, MobileNetV3Small achieved a superior balance of efficiency and accuracy, attaining a validation accuracy of 97.92%.

Additionally, the proposed model demonstrated a high recall value of 98.72%, which is critical in ensuring that actual faulty panels are not misclassified as normal. The precision of 97.89% indicates that the model effectively minimizes false positives, avoiding unnecessary maintenance costs. Additionally, a higher F1 score (98.24%) and MCC score (97.44%) confirm that the model can ensure balanced performance, even in the presence of class imbalances. Thus, the MobileNetV3Small based model provided consistent performance across all fault categories.

In contrast, models like NASNetMobile and InceptionV3 exhibited the lowest validation performance with an accuracy of only 55.21% and 62.50%. This implies that these models are inadequate in accurately classifying and capturing the distinctive characteristics found in images of solar panels. Overall, the proposed MobileNetV3Small model surpasses these existing architectures by maintaining high performance across all key metrics, confirming its robustness and reliability.

7.2.3. Performance on test set

Evaluation of unseen test sets is necessary to assess how much a model is generalized and not biased to a specific scenario. Table 16 shows how well the proposed approach performed on the test set in comparison to other classification models. MobileNetV3Small dominated regarding performance on the test set. It reached an accuracy of 97.53%, an F1 score of 97.74%, precision of 97.74%, recall of 97.74%, and MCC of 96.93%. These results bring out the robustness of MobileNetV3Small when deployed over previously unseen data. This is essential for deployment in real-world solar panel monitoring systems.

On the test set, other models, such as ConvNextXlarge, ResNet152, and EfficientNetV2B3 also showed noticeable performance. The EfficientNetV2B3 achieved 92.59% accuracy following the performance of

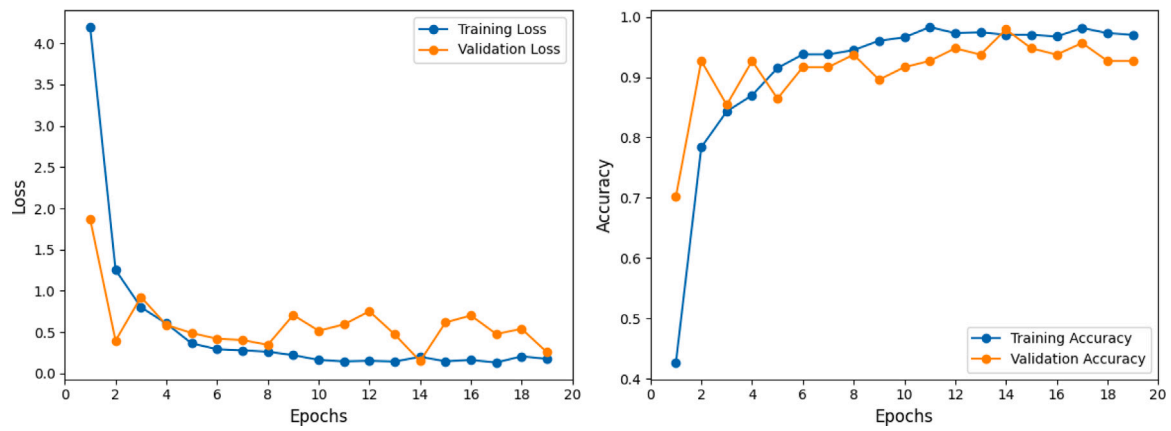


Fig. 11. Loss and accuracy curve of the classification model.

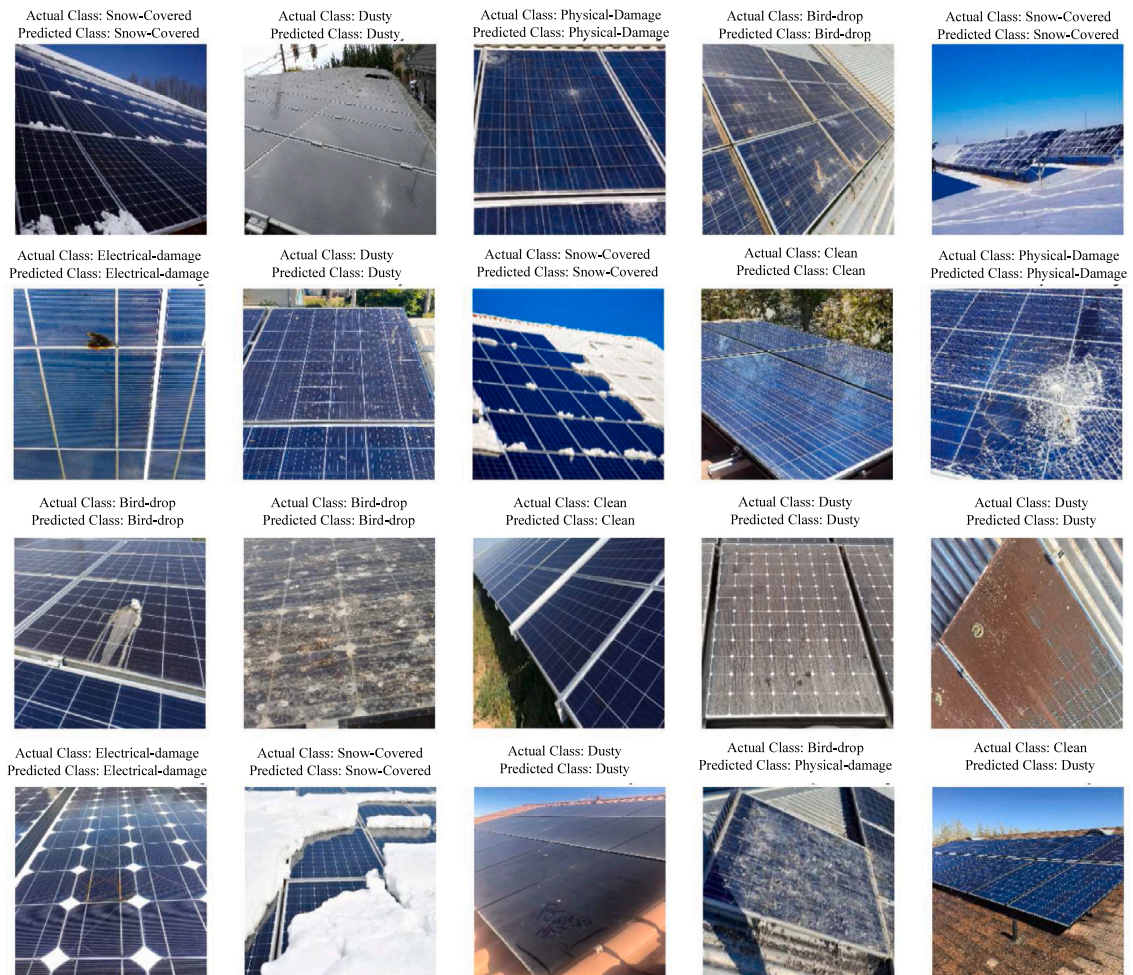


Fig. 12. Classification performance for different test samples.

MobileNetV3Small, ResNet152 and ConvNextXlarge also maintained a high accuracy value of 91.36% and 89.95%, but these models are less efficient and computationally heavy. Nonetheless, the comparison of these models' efficiency and performance indicates that the proposed model is more suitable for classifying solar panel faults and anomalies.

7.2.4. Analysis of model suitability

Fig. 12 demonstrates the actual class and predicted class of a few samples of the test set. Figs. 13 and 14 display the confusion matrices for the test and validation sets, respectively. These figures illustrate the models' classification performance across several classes.

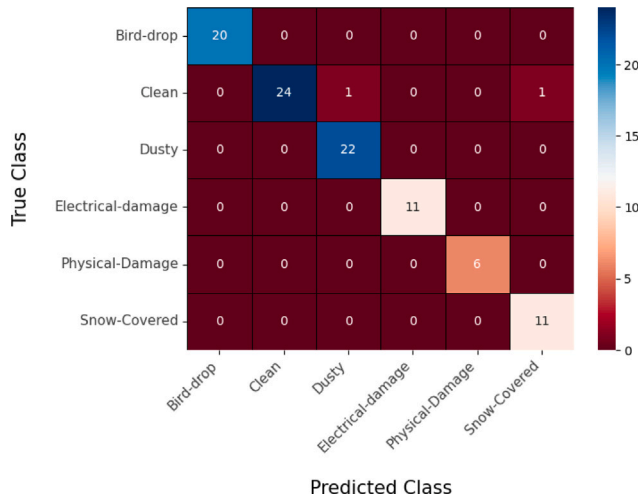


Fig. 13. Confusion matrix for the validation set.

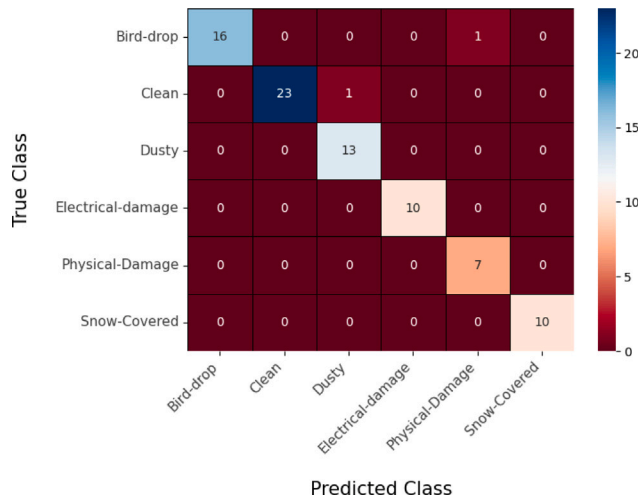


Fig. 14. Confusion matrix for the test set.

Overall, the excellent performance of this model across both test and validation sets demonstrates an effective balance between complexity and computational efficiency. Because of this, MobileNetV3Small is especially well-suited for resource-constrained scenarios where high precision and effective model deployment are crucial, like monitoring systems set up at distant solar farms.

7.2.5. Performance on additional datasets

To assess the robustness of the classification module beyond the primary dataset, it was evaluated on the SolNet [31] and Solar Panel dust detection [41] datasets. These datasets contained mainly two classes, one is Clean and another is Dusty.

As Table 17 shows, on SolNet [31], the model achieved an accuracy of 97.22% and 97% F1-score. While on Solar Panel dust detection [41] dataset, it achieved an accuracy of 95.53% and an F1 score of 94.53%. The high performance for different datasets demonstrates that the fault classification module has not overfit to a specific dataset and can be adapted to different PV module imaging conditions.

7.3. Computational efficiency and deployment analysis

The computational efficiency of the proposed framework was evaluated for both the segmentation and classification modules, focusing

Table 17

Classification performance on additional datasets.

Model	Accuracy	Precision	Recall	F1	MCC
SolNet [31]	97.22	97	97	97	93.98
Solar panel dust detection [41]	95.53	93.14	95.24	94.53	93.2

Table 18

Computational analysis of the proposed framework.

Module	Dataset	Inference time (ms)	FPS	Model size (MB)
Segmentation	PV01	75.2	13.3	10.34
	PV03	101.4	9.86	
Classification	Clean faulty	56.18	17.8	5.85

Table 19

Performance analysis on PV01 for input resolutions.

Input size	IoU	Accuracy	Dice	Recall	Precision	F1
128 × 128	90.60	97.35	95.06	95.60	96.07	95.82
244 × 244	92.94	97.70	96.34	95.94	97.70	96.81
256 × 256	93.63	98.31	96.7	97.45	97.04	97.24
512 × 512	91.30	97.67	95.45	95.45	97.30	96.36

Table 20

Performance analysis on PV03 for input resolutions.

Input size	IoU	Accuracy	Dice	Recall	Precision	F1
128 × 128	88.27	97.24	93.72	95.59	94.33	94.91
244 × 244	89.04	97.63	94.15	95.07	95.89	95.44
256 × 256	91.24	98.06	95.37	95.02	96.95	95.93
512 × 512	90.88	97.44	95.16	94.92	96.96	95.88

on three key aspects: inference time per image, achievable frames per second (FPS), and overall model size. The results are summarized in Table 18.

The segmentation module demonstrated an inference time of 75.2 ms per image (13.3 FPS) on the PV01 dataset and 101.4 ms per image (9.86 FPS) on PV03. The slight increase in processing time for PV03 can be attributed to increased variability and complexity in image backgrounds. Despite this, the achieved FPS values confirm that the module can operate at near-real-time speeds, which is critical for applications such as live drone-based PV inspection.

The classification module achieved a faster inference time of 56.18 ms per image (17.8 FPS). With a compact size of 5.85 MB, the classification model is highly suitable for embedded deployment. Similarly, the segmentation model's size of 10.34 MB allows it to fit comfortably within the storage and memory constraints of modern edge devices such as NVIDIA Jetson Nano or even high-end Raspberry Pi boards.

Different input resolution. To assess the model's sensitivity to input resolution, it has been trained with four different image sizes as presented in Tables 19 and 20. On PV01, the 256 × 256 configuration delivered a balance of performance and computational efficiency, achieving 93.63% IoU and 97.24% F1-score. Notably, while 512 × 512 yielded slightly lower IoU (91.30%) and increased computational cost, the difference was small. Even for lower input resolution, the IOU difference was only 2% to 3%. It indicates that the model can adapt well to higher resolutions. Similarly, for the PV03 dataset, the module provided balanced performance for different input resolutions. For the different input sizes, the variation in IOU was a maximum of 1% to 3%. This highlights the robustness of the proposed model to input resolution variations. The ability to maintain high performance across multiple input sizes suggests that the proposed segmentation framework can be flexibly deployed in diverse operational contexts without significant loss of accuracy.

Table 21

Performance analysis of the segmentation module on PV01 dataset for different training data sizes.

Train:Val:Test	IoU	Accuracy	Dice	Recall	Precision	F1
80:10:10	93.63	98.31	96.7	97.45	97.04	97.24
70:15:15	92.74	97.42	95.76	95.71	96.55	96.12
60:20:20	90.89	97.70	95.19	96.01	96.00	95.99

Table 22

Performance analysis of the segmentation module on PV03 dataset for different training data sizes.

Train:Val:Test	IoU	Accuracy	Dice	Recall	Precision	F1
80:10:10	91.24	98.06	95.37	95.02	96.95	95.93
70:15:15	89.18	97.15	94.15	95.34	94.58	94.84
60:20:20	87.92	96.95	93.45	95.98	93.44	94.62

Table 23

Performance analysis of the classification module for different training data sizes.

Train:Val:Test	Accuracy	Precision	Recall	F1	MCC
80:10:10	97.92	97.89	98.72	98.24	97.44
70:15:15	94.531	95.14	94.242	94.525	93.2
60:20:20	90.741	93.234	88.18	90.17	92.96

Variation in training data size. Both the segmentation and classification modules were evaluated with varied training data sizes to investigate their efficiency and robustness. Tables 21 and 22 represent the performance of the RefineU-Net model with training data size variation for PV01 and PV03 datasets, respectively. For PV01, the highest overall metrics were obtained with the 80:10:10 configuration, achieving an IoU of 93.63%. Interestingly, reducing the training ratio to 70% did not result in a substantial drop in performance, with an IoU of 92.74%. Even with only 60% of the data used for training, the model maintained a competitive performance of 90.89% IoU. It demonstrates data efficiency and the capacity to learn robust segmentation features with limited annotated samples.

On PV03, the same trend was observed. While 80:10:10 yielded the best performance with 91.24% IoU, both 70% and 60% training ratios sustained high performance, with only moderate decreases in IoU. This resilience suggests that the model architecture, combined with effective regularization, enables reliable performance even in low-data regimes. It is an important property for real-world PV monitoring scenarios where labeled data may be scarce.

Similarly, the classification module's performance has been examined under varying training data sizes to determine its data efficiency. As presented in Table 23 with the 80% training split, the model achieved 97.92% accuracy and 98.24% F1-score, indicating strong fault recognition capabilities. Notably, performance remained high with reduced training data, 94.53% accuracy for 70:15:15 and 90.74% accuracy for 60:20:20 train, validation, and test split. Even in the lowest data regime, the model maintained a high MCC (92.96). It demonstrates the robustness and ability of the model to generalize well despite reduced training samples. This makes the module suitable for scenarios with limited annotated fault datasets.

From a deployment perspective, these characteristics make the proposed framework highly adaptable to resource-constrained and low-power environments, which are common in remote PV monitoring setups. The lightweight architecture minimizes computational burden, enabling continuous monitoring without requiring high-performance computing infrastructure.

The segmentation and classification modules together can sustain near real-time performance, making them viable for onboard processing in unmanned aerial vehicles (UAVs) or low-power devices commonly used in solar farms. For large-scale PV installations, scalability can be achieved by deploying multiple edge devices in parallel, though

cumulative latency may increase when processing continuous drone footage over very large areas.

The analysis of input resolutions further highlights the adaptability of the framework. While 256×256 provides the best trade-off, higher-resolution imagery from drones or satellites substantially increases computational cost. In such scenarios, image downscaling, tiling, or cloud-assisted computation may be necessary.

Overall, the framework demonstrates near real-time feasibility for PV inspection, robustness to input variations, and resilience under reduced training data. Despite there are some challenges, the proposed solution remains scalable and energy-efficient, making it well-suited for large-scale PV installations.

8. Explainable AI based interpretation

This section demonstrates interpretability for the predictions of the proposed MobileNetV3-Small-Based classification model.

8.1. LIME explanation

In this study, LIME was applied post hoc after the classification model was fully trained, rather than being integrated into the training or inference pipeline. The objective was to analyze the decision-making process of the MobileNetV3Small-based classifier across all six solar panel fault categories. For each class, LIME-generated saliency maps were inspected to determine whether the regions of highest importance corresponded to known fault areas. Validation was also conducted through feedback from domain expert, who confirmed that the highlighted regions generally aligned with physically meaningful features in most cases.

The regions of interest (ROIs) varied across the different classes, reflecting the specific characteristics of each abnormality or fault. For the "Physical-Damage" and "Electrical-Damage" classes, the ROI was confined to the visibly damaged areas of the solar panels. Whereas for the "Snow-Cover" class, the ROI was the snow on the panel. The "Bird-Drop" class targeted specific areas with bird droppings, while the "Dusty" class identified dusty patches on the solar panel. Superpixels within these ROIs were assigned higher weights by the LIME algorithm, indicating that they had a significant influence on the model's predictions.

Figs. 15 to 20 illustrate the LIME explanations for different classes. For each sample, the original image, along with LIME boundary visualizations, masked images, and heatmaps, has been represented. The LIME boundary and masked image highlight the specific regions that contribute most to the model's classification. Additionally, the heatmap provides a color-coded representation of feature importance, ranging from blue to red. For a given class, the darker blue areas indicate the most influential areas in the model's prediction. Red areas, on the other hand, represent areas that negatively influence or reduce the confidence of the prediction.

Fig. 15 highlights examples from the "Physical-Damage" class. For this class, the LIME explainer emphasized the physically damaged regions of the solar panel. The LIME explainer created a boundary around the damaged region. It is more visible in the Heatmap image that the classification model is giving more weight to the damaged area, as represented by a dark blue color. This clear visualization reinforces the model's reliance on the damaged area for its predictions.

Similarly, Fig. 16 presents samples from the "Electrical-Damage" class, where LIME highlighted areas exhibiting electrical faults. In both samples, the boundaries generated by LIME validate that the classification model prioritized the superpixels related to the electrical damage area. This capacity for interpretability allows practitioners to verify that the model's focus aligns with expert knowledge about solar panel faults.

The "Bird-Drop" class is depicted in Fig. 17, where LIME explanations reveal that the model identifies relevant areas affected by

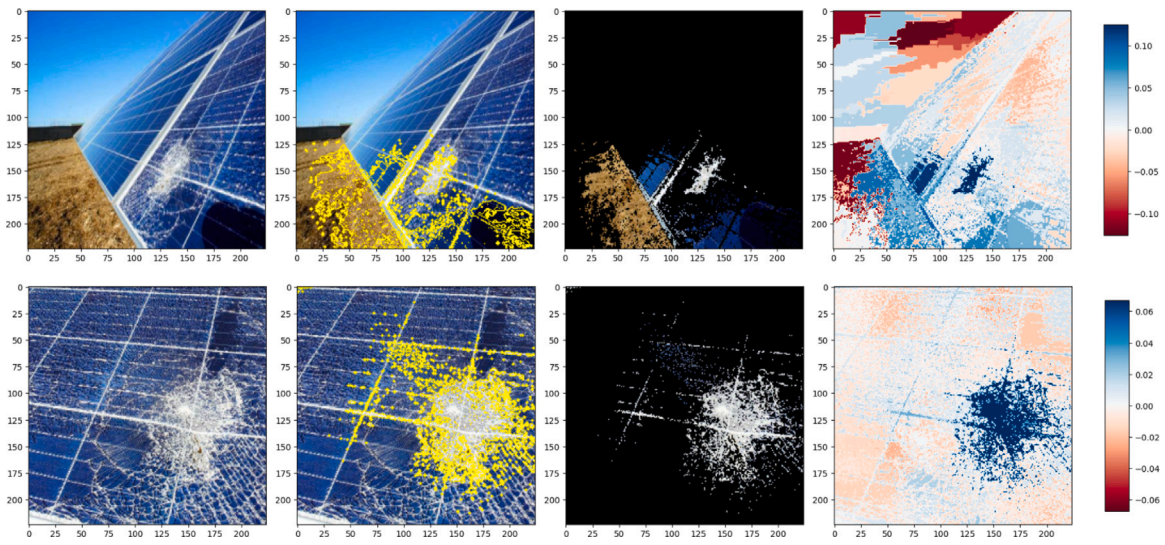


Fig. 15. LIME explanations for Physical-Damage samples (Original image, boundary visualization, masked image, and heatmap).

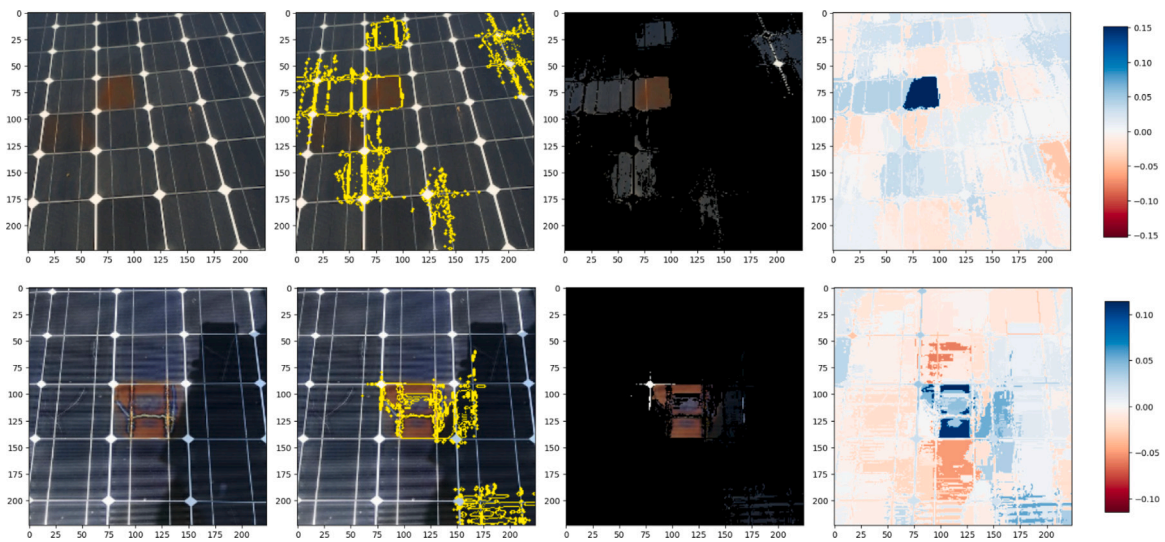


Fig. 16. LIME explanations for Electrical-damage samples (Original image, boundary visualization, masked image, and heatmap).

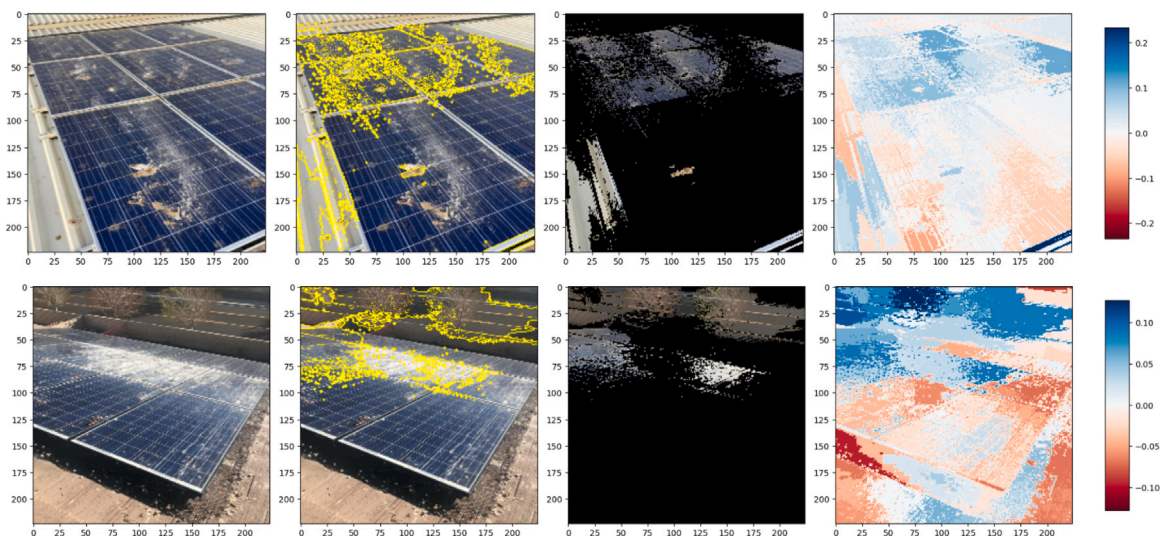


Fig. 17. LIME explanations for Bird-drop samples (Original image, boundary visualization, masked image, and heatmap).

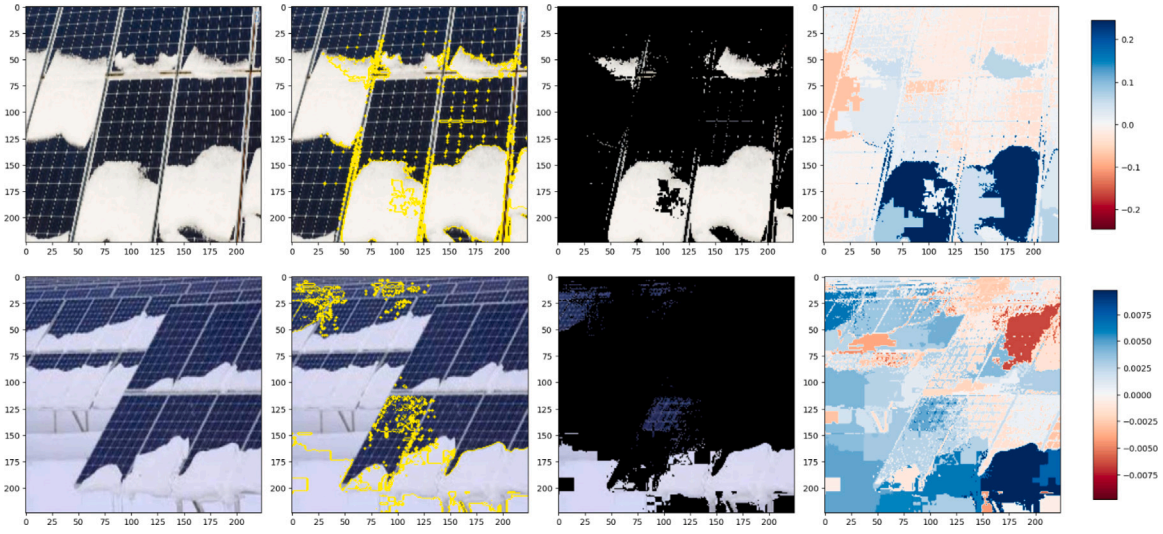


Fig. 18. LIME explanations for Snow-Covered samples (Original image, boundary visualization, masked image, and heatmap).

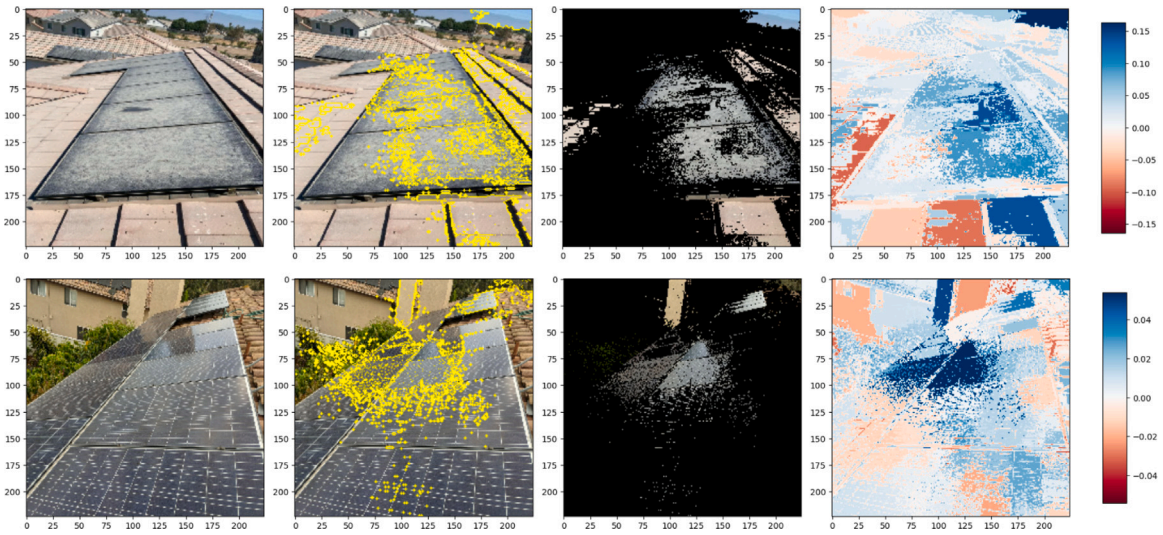


Fig. 19. LIME explanations for Dusty samples (Original image, boundary visualization, masked image, and heatmap).

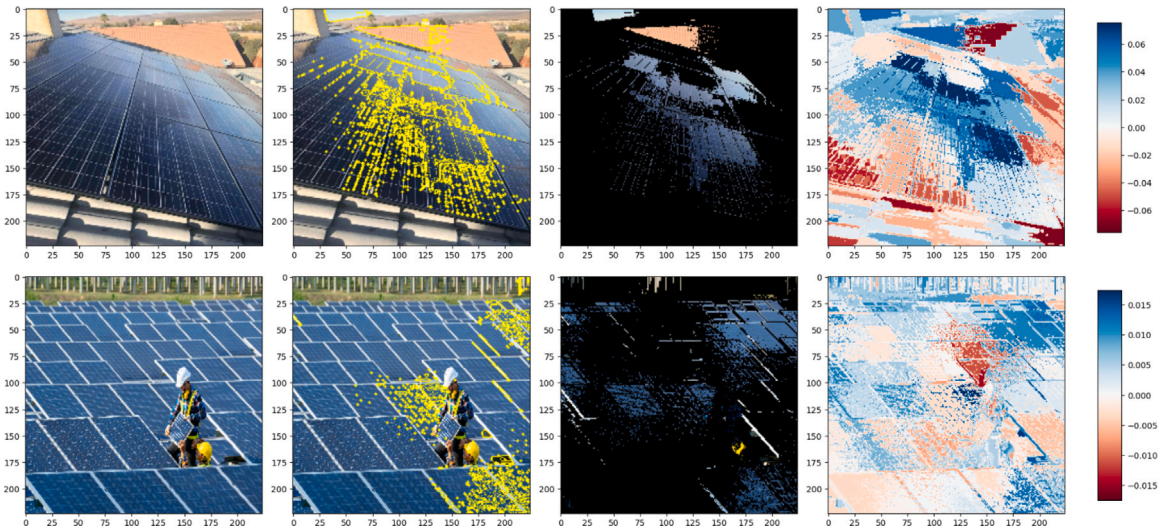


Fig. 20. LIME Explanations for Clean samples (Original image, boundary visualization, masked image, and heatmap).

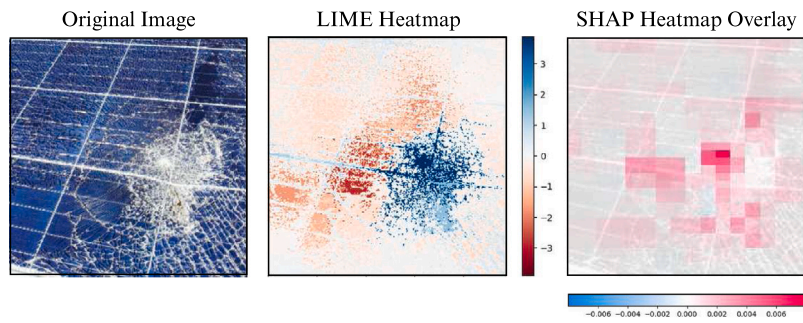


Fig. 21. Comparison of LIME and SHAP explanations for Physical-Damage samples (Original image, LIME heatmap, SHAP heatmap overlay).

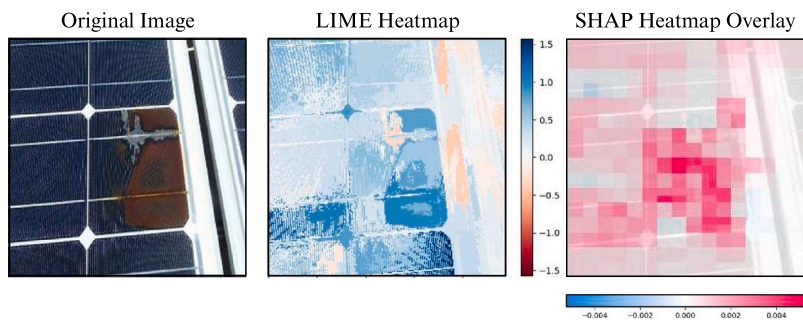


Fig. 22. Comparison of LIME and SHAP explanations for Electrical-damage samples (Original image, LIME heatmap, SHAP heatmap overlay).

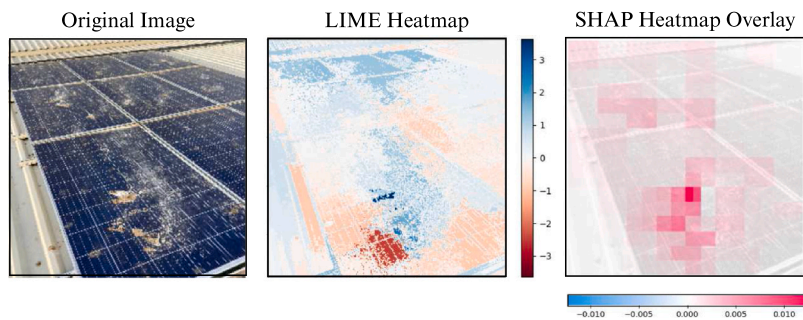


Fig. 23. Comparison of LIME and SHAP explanations for Bird-drop samples (Original image, LIME heatmap, SHAP heatmap overlay).

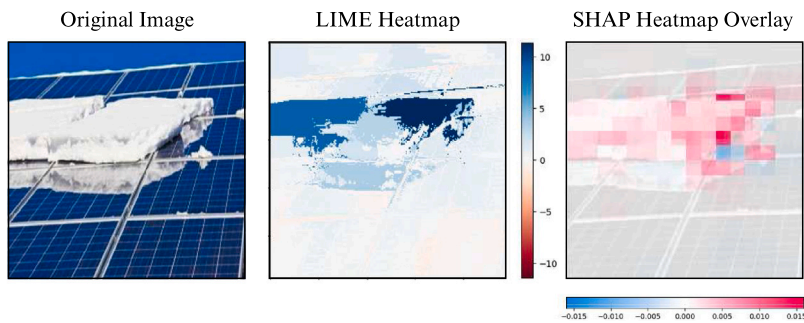


Fig. 24. Comparison of LIME and SHAP explanations for Snow-Covered samples (Original image, LIME heatmap, SHAP heatmap overlay).

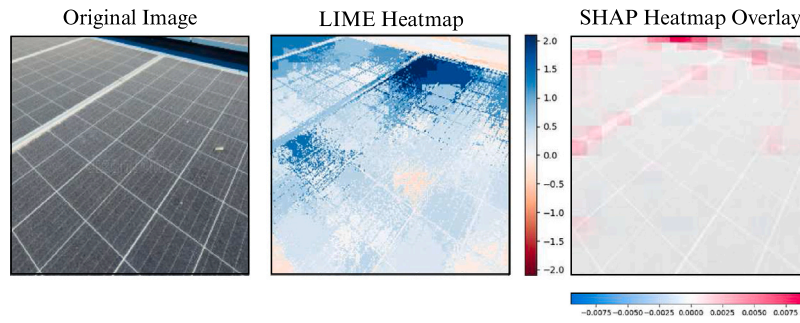


Fig. 25. Comparison of LIME and SHAP explanations for Dusty samples (Original image, LIME heatmap, SHAP heatmap overlay).

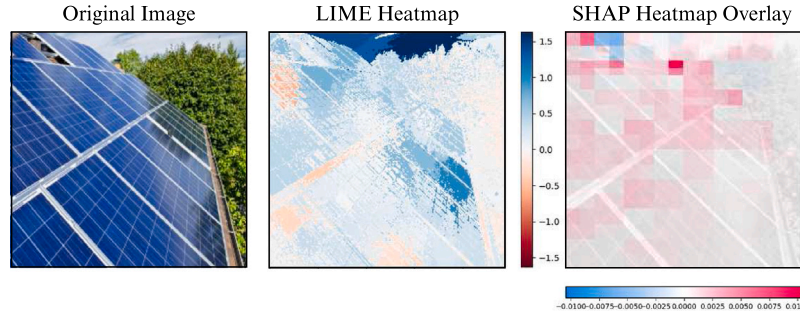


Fig. 26. Comparison of LIME and SHAP explanations for Clean samples (Original image, LIME heatmap, SHAP heatmap overlay).

bird droppings without needing to cover all affected regions. This observation demonstrates the model's ability to generalize to partial occlusion, with LIME pinpointing only the essential superpixels that drive the classification.

This capacity to categorize a sample based on partial information demonstrates the model's robustness in real-world applications. Although in a few samples, LIME highlighted irrelevant regions (e.g., stains not caused by birds), showing potential inconsistencies in explanation quality.

In Fig. 18, examples from the "Snow-Cover" class show the expected ROI as the snow on the panel represented a significant area in the heatmap. However, for the second sample, the model is also making boundaries around the snow that is not on the solar panel. In this case, the LIME explainer indicates that the incorrect superpixels are also being prioritized for the sample classification. Thus, it offers insights into areas where the model may require further refinement.

For the "Dusty" class, as seen in Fig. 19, LIME highlights the dusty areas of the solar panel. The model classified a solar panel as "Dusty" even with minimal dust coverage, emphasizing its sensitivity to such features. This behavior suggests that while the model is effective in identifying dust on panels, it may also lead to false positives if not calibrated correctly.

For the "Clean" class, the behavior of the model is totally different, as represented in Fig. 20. Like other classes, the "Clean" class sample does not have any particular ROI, such as a damaged area or snow. The absence of defining features of other classes leads the model to classify the panel as "Clean". This observation highlights that a "clean" classification relies on the absence of faults rather than the identification of specific positive attributes.

Overall, the insights facilitated by LIME underscore its critical role in interpreting the regions influencing model predictions. Thus, the reliability and transparency of classification outcomes are validated across different fault and abnormality classes in solar panels.

8.2. Comparative analysis of LIME and SHAP explanations

To enhance interpretability and validate the explanations produced by LIME, SHAP was introduced as a complementary XAI technique.

Figs. 21 to 26 illustrate a side-by-side comparison of LIME and SHAP outputs across the six fault categories. Each figure presents the original image, a LIME heatmap, and a SHAP heatmap overlay. In SHAP visualizations, red regions denote features that positively contribute to the predicted class (increasing confidence), while blue regions indicate features that negatively influence the prediction. The intensity of color corresponds to the magnitude of contribution, with darker shades reflecting stronger influence. LIME heatmaps follow a similar principle, where darker blue areas indicate stronger positive influence and red areas indicate negative influence.

For the "Physical-Damage" case (Fig. 21), LIME sharply focused on the cracked or broken portions of the panel, confirming the model's reliance on visible structural defects. SHAP also highlighted these damaged zones in red but attributed some importance to surrounding areas, possibly due to textural or shading variations linked to the damage.

In the "Electrical-Damage" case represented in Fig. 22, LIME localized superpixels directly over the burn marks, closely aligning with the defect. SHAP similarly highlighted these regions in red but extended coverage to adjacent zones, suggesting attention to indirect contextual cues.

For the "Bird-Drop" sample in Fig. 23, both LIME and SHAP emphasized nearly identical areas corresponding to the droppings, reinforcing the model's prediction. In the "Snow-Cover" case presented in Fig. 24, both methods prioritized snow-covered regions, though LIME highlighted the entire snow patch with dark blue, while SHAP concentrated on portions of it.

For the "Dusty" panel, the LIME explanation is comparatively better than SHAP as presented in Fig. 25. LIME sharply outlined dusty patches, even when coverage was minimal, showing high sensitivity to dust presence. LIME heatmap highlighted the whole dusty panel with blue, indicating its important in model prediction. Whereas the SHAP heatmap only highlighted a few edges as dusty panel, which may misinterpret the prediction process in some cases.

For "Clean" samples, LIME generally produced no strongly bound regions of interest, reflecting the absence of fault features as illustrated in Fig. 26. SHAP displayed a similar explanation for the "Clean" label.

Overall, both techniques effectively captured salient fault features across classes. However, minor inconsistencies arose, particularly in

cluttered or partially occluded images, where background regions not related to faults were sometimes emphasized. These observations indicate that while SHAP and LIME are complementary, LIME offered sharper localization in certain cases.

9. Conclusion

This study presents a novel deep learning based framework for solar panel segmentation and fault classification, incorporating explainable AI to enhance model interpretability. The proposed hybrid RefineU-Net model demonstrates superior segmentation performance, accurately delineating the boundaries of solar panels. The MobileNetV3Small-based model can accurately classify various faults and abnormalities. Furthermore, the transparency of model predictions is ensured by incorporating LIME-based explanations. The following enumeration showcases the noteworthy findings of this work:

- The hybrid RefineU-Net segmentation model effectively detected solar panel regions, outperforming traditional segmentation methods like U-Net, Deeplabv3+, and RefineNet. It performed better compared to the existing research with an IoU of 93.63% and 91.24% for two different datasets.
- For fault and abnormality classification, the MobileNetV3Small-based transfer learning model achieved an accuracy of 97.92% and 97.53% for the validation and test set, respectively. Thus, it efficiently identified various real-world faults and abnormalities, including physical damage, electrical damage, dust accumulation, bird droppings, and snow cover.
- By integrating LIME-based explainable AI, the framework enhanced model transparency, ensuring that decisions are interpretable and aligned with domain knowledge.

However, there are some key limitations of this study, and further improvements can be explored. Future work should focus on incorporating additional fault or abnormality types and analyzing fault progression over time. Implementing the approach on a larger and more diverse dataset can improve its overall generalizability. More recent AI models can be explored to build a framework that is more computationally efficient. The integration of other XAI techniques, such as Grad-CAM and Grad-CAM++, can be explored for further enhancement.

In conclusion, the proposed framework represents a significant advancement toward intelligent and reliable solar panel monitoring systems, facilitating smarter, data-driven maintenance and maximizing energy efficiency in renewable energy applications.

CRedit authorship contribution statement

Asif Ur Rahman Adib: Writing – original draft, Software, Methodology, Conceptualization. **Mainul Islam:** Writing – original draft, Resources, Formal analysis. **Md. Shadman Abid:** Writing – review & editing, Validation, Methodology, Conceptualization, Software. **Razzaqul Ahshan:** Writing – review & editing, Supervision, Resources, Project administration.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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